Li-Fi Link Using BFSK: Complete Project Kit

Final Year Project, B.Tech ECE, NIT Puducherry

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1 November 2026

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Part A: Plan, Ideas, Learning and Schedule

1. The idea, formulated

Working title: Design and Implementation of a Low-Cost, Ambient-Light-Robust Visible Light Communication Link Using BFSK Modulation

Problem. Radio spectrum is crowded and RF is unwanted in some places (hospitals, aircraft cabins, mines, underwater, RF-sensitive labs). Visible light is unlicensed, does not pass through walls (a natural security boundary) and every LED is already a potential transmitter. The usual student Li-Fi demo uses on-off keying (OOK), which breaks easily under sunlight, tube lights and 100 Hz mains flicker.

Our approach. Encode bits as two tones (f0 for 0, f1 for 1) on the LED current. Since the information lives in *frequency*, not brightness, the receiver can high-pass and band-pass away ambient DC light and 100 Hz flicker, and the LED's average brightness stays constant (no visible flicker). Demodulate two ways and compare: (a) the classic XR2211 PLL FSK demodulator IC, (b) a Goertzel-filter demodulator running on an ESP32.

Objectives 1. Build a working one-way VLC link: text, then files, over 0.5 to 3 m at 1 to 10 kbps. 2. Achieve BER below 1e-3 at 1 m under normal room lighting. 3. Compare PLL (XR2211) vs software Goertzel demodulation on BER, range, cost and complexity. 4. Characterise BER vs distance, angle, ambient light (dark, tube light, sunlight) and bit rate. 5. Add framing + CRC-16 (and Hamming(7,4) FEC) and measure its effect.

Expected contribution (the "research gap" angle). A reproducible, sub Rs 3,000 BFSK VLC prototype with a side-by-side, measured comparison of hardware PLL and MCU Goertzel demodulation under real Indian indoor lighting (CFL/LED tube flicker at 100 Hz), plus an adaptive tone-selection scheme. That is enough for a conference paper.

2. Novelty ideas (pick 1-2 on top of the core)

#	Idea	Novelty	Feasibility	Demo impact
1	**Adaptive tone selection**: RX measures ambient spectrum, tells TX (via IR or BLE back-channel) which f0/f1 pair is cleanest	High	Medium	High
2	**PLL vs Goertzel head-to-head** with BER curves	Medium	High	Medium (paper-friendly)
3	**Bidirectional link**: visible downlink + IR uplink (true Li-Fi, not just VLC)	Medium	Medium	High
4	**Audio streaming** over light (play music via torch beam)	Low	High	Very high (crowd demo)
5	**Hamming(7,4)/CRC + ARQ** reliable file transfer	Medium	High	Medium
6	**Multi-FSK (4-FSK)** to double rate	Medium	Medium	Medium

#	Idea	Novelty	Feasibility	Demo impact
7	**Li-Fi indoor positioning**: each ceiling LED sends its ID as an FSK tone pair, phone/receiver locates itself	High	Medium	High
8	**Smartphone receiver**: decode FSK from the phone's ambient light sensor or camera rolling shutter	High	Low-Medium	Very high
9	**Secure short-range data exchange** (light-confined key exchange for pairing IoT devices)	High	Medium	Medium
10	**Underwater or vehicle-to-vehicle** demo (tail-light to car) in a water tank / toy cars	Medium	Medium	Very high
11	**Solar-panel receiver**: use a small solar cell as both energy harvester and photodetector	High	Medium	High

Recommendation: Core BFSK link + Idea 2 (comparison, makes the paper) + Idea 1 or 11 (the novelty) + Idea 4 for the demo day wow factor.

3. What you need to learn (in order)

1. **Digital comm basics**: ASK/FSK/PSK, coherent vs non-coherent detection, BER, Eb/N0. (NPTEL "Principles of Digital Communication", Lathi/Haykin chapters on FSK.)
2. **Optics for VLC**: LED electro-optical response and bandwidth, photodiode responsivity, Lambertian channel model, line-of-sight path loss. (Haas "What is LiFi"; Komine & Nakagawa 2004.)
3. **Analog front end**: transimpedance amplifier design (gain, Cf for stability, noise), active filters (Sallen-Key, MFB bandpass), comparators. (TI "Transimpedance Considerations" app notes; TI Filter Designer; Analog Filter Wizard.)
4. **PLL & FSK ICs**: XR2211 datasheet formulas, LM567 tone decoder.
5. **DSP**: sampling, Goertzel algorithm, windowing, bit-timing recovery.
6. **ESP32 firmware**: LEDC PWM, ADC continuous / I2S mode, timers and ISRs, FreeRTOS tasks (also builds your RTOS skill for job applications).
7. **Tools**: LTspice (simulate RX), KiCad (PCB), Python (numpy, matplotlib, pyserial) for simulation and logging.
8. **Writing**: IEEE paper format (Overleaf IEEE template), how to plot BER curves properly.

This also ticks Pratham's skill path: RTOS on ESP32 and a real fabricated KiCad PCB.

4. Schedule (fits your semester)

Conference target: IEEE CONECCT 2027 (Bengaluru, 29 to 31 Jul 2027) is the best timing fit; check its paper deadline (usually spring).

Calendar anchors: FA-Theory Nov 23 to Dec 3 2026; vacation from Dec 7; even sem from Jan 4 2027; GATE in Feb 2027; internship from mid-May 2027.

Phase	Dates	Work	Deliverable
0. Learn + read	Nov 1 to Nov 20 2026 (light, around exams)	Read 6 key papers, digital comm FSK refresh, run the Python simulation	Notes, sim plots
Exams	Nov 21 to Dec 6 2026	Pause	
0b. Order XR2211 (import)	This week, Nov 2026	XR2211 is obsolete and not stocked in India: order from an overseas seller now, or plan two LM567s as fallback	Part on the way
1. Order + breadboard TX	Dec 7 to Dec 20 2026	Order BOM, LED driver, ESP32 LEDC tone TX, check tones on scope	TX working
2. RX front end	Dec 21 2026 to Jan 3 2027	Photodiode + TIA + filters on breadboard, LTspice matched to scope	Clean tone at RX
3. Demodulation	Jan 4 to Jan 31 2027	XR2211 path + ESP32 Goertzel path, first text over light	First end-to-end text
GATE prep	Feb 2027	Minimal project work	
4. Framing + reliability	Mar 1 to Mar 21 2027	Preamble/sync, CRC-16, Hamming(7,4), PC tool	File transfer
5. Experiments	Mar 22 to Apr 18 2027	BER vs distance/angle/ambient/bit-rate, PLL vs Goertzel	Data + plots
6. Novelty	Apr 19 to May 10 2027	Adaptive tones or solar-cell RX, audio demo	Feature demo
Internship	mid-May to Jul 2027	Paper draft in spare time	Draft paper
7. PCB	Aug 2027	KiCad TX and RX boards, fabricate (JLCPCB/PCBPower), assemble	Working PCBs
8. Paper + review 1	Sep to Oct 2027	Submit to a conference, project review 1	Paper submitted
9. Polish + report	Nov 2027 to Mar 2028	Enclosure, final report, demo video, GitHub	Final report, viva

If your college's final project timeline is different, shift phases 7 to 9; phases 0 to 6 are the real build and should be done early so the final year is just polish and paper.

5. Further work / extensions after the core

- Raise the rate: faster photodiode (BPW34 -> SFH 213 / PIN with lower capacitance), dedicated LED driver, move to 4-FSK or OFDM.
- Full-duplex networking: multiple users (FDMA by tone pairs), a tiny Li-Fi access point.
- Integration with an IoT gateway (MQTT over light).
- Camera/smartphone receiver app.
- Underwater VLC with blue LEDs.
- Compare against IEEE 802.15.7 PHY I modes.

6. Report outline (final year report)

1. Introduction and motivation
2. Literature survey
3. Theory: VLC channel, BFSK, PLL and Goertzel demodulation
4. System design: TX, RX, calculations
5. Firmware and software
6. Simulation results
7. Hardware results and experiments
8. Comparison and discussion
9. Conclusion and future work References, appendices (schematics, BOM, code)

7. Viva questions to prepare

- Why BFSK and not OOK? What is the trade-off in bandwidth?
- How does your RX reject 100 Hz flicker and sunlight?
- What limits your data rate (LED bandwidth, photodiode capacitance, TIA GBW, MCU sampling)?
- Explain Goertzel vs FFT. Why Goertzel here?
- XR2211 timing: how did you choose R0, C0, R1?
- Theoretical BER of non-coherent BFSK? How does your measured curve compare?
- Eye safety and flicker: is your LED within safe limits?

Part B: Research Pack

Research Pack: Li-Fi / VLC Link Using BFSK Modulation

Project: LED transmitter switching between two tones (f0 for bit 0, f1 for bit 1) -> photodiode + TIA -> filtering -> BFSK demodulation (XR2211 / LM567 PLL, or Goertzel on ESP32). **Prepared for:** B.Tech ECE final-year project, NIT Puducherry. Compiled 31 Oct 2026 (IST). **Rule used:** every URL below was either opened (fetched) or returned directly in a search result during this research. Items marked *(abstract/metadata only)* were checked at the publisher/DOI page but the full text was not read. Items marked *(cited, not opened)* were seen only as a DOI inside another paper's reference list.

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1. Literature (27 sources)

(*) = one of the **6 closest to a BFSK student design** (summary list at the end of this section).

A. Standards 1. IEEE Std 802.15.7-2018, *IEEE Standard for Local and Metropolitan Area Networks, Part 15.7: Short-Range Optical Wireless Communications*, IEEE, 2019 (approved Dec 2018). DOI 10.1109/IEEESTD.2019.8697198 URL: https://standards.ieee.org/ieee/802.15.7/6820 · https://ieeexplore.ieee.org/document/8697198 (listed as Free Access on Xplore) Summary: Defines PHY and MAC for short-range optical wireless (190 nm to 10,000 nm), including ambient-light noise, dimming and flicker requirements and optical camera communication. Original 2011 version: https://standards.ieee.org/ieee/802.15.7/5154/ Borrow: Frame structure ideas (preamble/sync header, length field, CRC), flicker and dimming vocabulary, and the PHY I (low-rate, outdoor-robust) mindset to justify a low-rate, noise-robust BFSK link. PHY I spans roughly 11.67 to 266.6 kb/s and PHY II 1.25 to 96 Mb/s (figures as quoted in the Photonics 2023 survey, https://doi.org/10.3390/photonics10090993).

2. S. Rajagopal, R. D. Roberts, S.-K. Lim, "IEEE 802.15.7 visible light communication: modulation schemes and dimming support," *IEEE Communications Magazine*, vol. 50, no. 3, pp. 72-82, Mar. 2012. DOI 10.1109/MCOM.2012.6163585 URL: https://ieeexplore.ieee.org/document/6163585/ · ETRI copy page: https://ksp.etri.re.kr/ksp/article/read?id=91 *(abstract/metadata only)* Summary: The standard authors' own explanation of OOK, VPPM and CSK in 802.15.7, and how dimming and flicker-free operation are achieved. Borrow: The "why not plain OOK" argument (flicker/dimming), and a ready-made table to compare against in your report.

3. IEEE Std 802.11bb-2023, *Wireless LAN MAC and PHY, Amendment 6: Light Communications*, IEEE, Nov 2023. DOI 10.1109/IEEESTD.2024.10315104 URL: https://standards.ieee.org/ieee/802.11bb/10823 ·

https://ieeexplore.ieee.org/document/10315104 Summary: Brings Li-Fi into the Wi-Fi family, operating in the 800-1000 nm band with 10 Mb/s to 9.6 Gb/s at the MAC SAP. IEEE SA now lists it as "Superseded" (folded into the 802.11 base). Borrow: One paragraph in your intro showing where commercial Li-Fi is heading (infrared, Wi-Fi-compatible), which contrasts nicely with a low-rate visible-light BFSK link.

B. Key surveys and overviews **4. H. Haas, L. Yin, Y. Wang, C. Chen**, "What is LiFi?," *Journal of Lightwave Technology*, vol. 34, no. 6, pp. 1533-1544, Mar. 2016. DOI 10.1109/JLT.2015.2510021 URL: https://ieeexplore.ieee.org/document/7360112 · open repository page: https://www.research.ed.ac.uk/en/publications/what-is-lifi-2/ Summary: Separates VLC (point-to-point link) from LiFi (fully networked, bidirectional, multiuser "attocells"), and covers components through to hybrid LiFi/WiFi networks. Borrow: Correct terminology. Your project is technically a **VLC link**, so call it "VLC/Li-Fi link" in the title and explain the difference in one paragraph.

5. P. H. Pathak, X. Feng, P. Hu, P. Mohapatra, "Visible Light Communication, Networking, and Sensing: A Survey, Potential and Challenges," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2047-2077, 2015. DOI 10.1109/COMST.2015.2476474 URL: https://ieeexplore.ieee.org/document/7239528 · free author PDF: http://phpathak.com/files/vlc-comsoccst.pdf Summary: Covers transmitter/receiver components, channel, modulation, MAC, programmable platforms and VLC sensing (localisation, gesture recognition). Borrow: The section on system design and programmable platforms, which is a good template for your "hardware architecture" chapter.

6. D. Karunatilaka, F. Zafar, V. Kalavally, R. Parthiban, "LED Based Indoor Visible Light Communications: State of the Art," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 3, pp. 1649-1678, 2015. DOI 10.1109/COMST.2015.2417576 URL: https://ieeexplore.ieee.org/document/7072557 · https://research.monash.edu/en/publications/led-based-indoor-visible-light-communications-state-of-the-art Summary: Indoor VLC from 1979 to 2014: VLC compared with IR and RF, modulation and dimming, filtering, and indoor challenges. Borrow: LED bandwidth limits (phosphor white LEDs are slow) and the dimming/modulation discussion for your literature-review chapter.

7. L. E. M. Matheus, A. B. Vieira, L. F. M. Vieira, M. A. M. Vieira, O. Gnawali, "Visible Light Communication: Concepts, Applications and Challenges," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 4, pp. 3204-3237, 2019. DOI 10.1109/COMST.2019.2913348 URL: https://ieeexplore.ieee.org/document/8698841 · free author PDF: https://www2.cs.uh.edu/~gnawali/papers/vlcsurvey-ieee19.pdf Summary: A more recent survey covering hardware, PHY/MAC, NOMA, applications and open challenges. It explicitly flags ambient light (sunlight, fluorescent, other LEDs) as a major deployment challenge. Borrow: The "open challenges" section supplies a citation for your problem statement on ambient-light interference.

8. J. M. Kahn, J. R. Barry, "Wireless infrared communications," *Proceedings of the IEEE*, vol. 85, no. 2, pp. 265-298, 1997. DOI 10.1109/5.554222 *(cited, not opened; DOI taken from the reference list of source 12)* URL: https://doi.org/10.1109/5.554222 Summary: The classic IM/DD optical-wireless channel paper (LOS/diffuse channel, shot noise from ambient light, modulation power efficiency). Most VLC channel models still build on it. Borrow: The Lambertian LOS channel gain equation for predicting received power against distance in your link-budget section.

C. FSK / BFSK in VLC (core of your project) **9. (*) C. Beguni, A. Done, A.-M. Căilean, S.-A. Avătămăniţei, E. Zadobrischi, **"Experimental Demonstration of a Visible Light Communications System Based on Binary Frequency-Shift Keying Modulation: A New Step toward Improved Noise Resilience," *Sensors*, vol. 23, no. 11, 5001, 2023. DOI 10.3390/s23115001 (open access,** full text read) **URL: https://www.mdpi.com/1424-8220/23/11/5001 Summary: A hardware BFSK VLC prototype (0.96/1.04 MHz tones, 10 kb/s NRZ, red XP-E2 LED, PIN-photodiode TIA, KA3361 FSK demodulator, Schmitt trigger, ARM M7). It was compared directly against OOK-Manchester on the same platform. BFSK tolerated about 25% more direct optical noise (3500 vs 2800 μW/cm^2) and about 20% more indirect noise, with BER $< 10^{-6}$. The range reached 26 m in lab conditions. Borrow:** This is your reference design.** Borrow the experimental method (calibrate OOK and BFSK to the same irradiance and range, then ramp a halogen noise source until BER $> 10^{-6}$), the block diagram, the Carson's-rule bandwidth reasoning, and the honest limitations (data rate, jitter, calibration difficulty). The paper also notes that AC incandescent lamps produce 100 Hz interference with harmonics up to about 2 kHz, and fluorescent lamps

produce interference up to about 40 kHz (low amplitude to 1 MHz). That spectrum is what you choose your tones to avoid.

10. (*) A. W. Azim, A. Rullier, Y. Le Guennec, L. Ros, G. Maury, "Energy Efficient M-ary Frequency-Shift Keying-Based Modulation Techniques for Visible Light Communication," *IEEE Transactions on Cognitive Communications and Networking*, vol. 5, no. 4, pp. 1244-1256, Dec. 2019. DOI 10.1109/TCCN.2019.2940260 *(abstract/metadata only)* URL: https://ieeexplore.ieee.org/document/8830382 Summary: Proposes DC-FSK (DC offset) and U-FSK (unipolar, sign-flipped halves) to make FSK compatible with intensity modulation, with ML and suboptimal receivers. In simulation these schemes are more energy-efficient than M-PAM for low-rate IoT. Borrow: The theory section. Your square-wave or DC-biased-sine LED drive is "DC-FSK" in their terms, and their BER expressions give you a theoretical curve to plot next to measured BER.

11. (*) G. M. Mougoue Yamga, A. R. Ndjiongue, K. Ouahada, **"Low complexity clipped Frequency Shift Keying (FSK) for Visible Light Communications," *2018 IEEE 7th Int. Conf. on Adaptive Science & Technology (ICAST)*, Accra, Aug. 2018. DOI 10.1109/ICASTECH.2018.8506727 *(abstract/metadata only)* URL: https://ieeexplore.ieee.org/document/8506727 Summary: Asymmetric FSK (AFSK) for non-negative VLC signals. It targets low cost and low complexity, mitigates flicker and dimming for square and rectified-sine carriers, and in simulation outperforms OOK for selected sizes. Borrow: Justification that driving the LED with square-wave tones** (what an ESP32 LEDC pin naturally produces) is a legitimate FSK variant, plus flicker arguments.

12. (*) M. Kang, J. Wu, J. Li, Y. Zhou, Z. Zhao, Y. Zhou, "Prolonging The Range of Low-power Visible Light Communication Systems with M-FSK," *2024 IEEE 30th Int. Conf. on Parallel and Distributed Systems (ICPADS)*, pp. 536-543, 2024. DOI 10.1109/ICPADS63350.2024.00076 *(abstract/metadata only)* URL: https://www.computer.org/csdl/proceedings-article/icpads/2024/159600a536/22f0z3Q2Vva Summary: Sine-intensity M-FSK on 1 W/3 W LEDs with a gradient-threshold demodulator. Reaches 5.8 m (1 W) and 10 m (3 W), which the authors describe as about 2.5× the range of prior low-power work, and argues FSK resists ambient light better than intensity (OOK) modulation. Borrow: A recent (2024) citation proving FSK-VLC is still active research, plus a low-power-LED range benchmark to compare against.

13. (*) N. Rajagopal, P. Lazik, A. Rowe, **"Visual Light Landmarks for Mobile Devices," *Proc. ACM/IEEE IPSN 2014*, Carnegie Mellon University. URL (author PDF): https://users.ece.cmu.edu/~agr/resources/publicatio vlc.pdf *(abstract read)* Summary: LED lights broadcast FSK tones that phone cameras or light sensors detect. Operates above 2 kHz to avoid visible flicker, detects frequencies up to 8 kHz with 0.2 kHz channel separation, and uses this for indoor landmarks/positioning. Borrow: Tone-planning numbers (flicker floor about 2 kHz, channel spacing) and a ready positioning extension**: each lamp sends a unique f0/f1 pair or ID.

14. C.-W. Chow, R.-J. Shiu, Y.-C. Liu, X.-L. Liao, K.-H. Lin, Y.-C. Wang, Y.-Y. Chen, "Using advertisement light-panel and CMOS image sensor with frequency-shift-keying for visible light communication," *Optics Express*, vol. 26, no. 10, pp. 12530-12535, 2018. DOI 10.1364/OE.26.012530 *(abstract only)* URL: https://pubmed.ncbi.nlm.nih.gov/29801291 Summary: An FSK VLC link from an advertising light panel to a smartphone camera using the rolling-shutter effect, with a success rate above 96% up to 200 cm. Borrow: An optional demo idea (phone camera as a second receiver) and evidence that FSK suits low-contrast optical channels.

15. "A Dimmable Offline LED Driver With OOK-M-FSK Modulation for VLC Applications," *IEEE Transactions on Industrial Electronics*, 2018. DOI 10.1109/TIE.2018.2868022 *(abstract only; authors not captured)* URL: https://doi.org/10.1109/TIE.2018.2868022 Summary: Embeds ON-OFF M-FSK into a single-stage PFC LED driver so more than one bit is sent per switching period without disturbing dimming. Borrow: The argument that FSK can coexist with real mains-powered LED drivers, useful for a "future work: integrate into a room luminaire" section.

16. "Comparative Study of Frequency-based Modulations for Visible Light Communication," *IEEE WCNC 2024*. DOI 10.1109/WCNC57260.2024.10571121 *(abstract only; authors not captured)* URL: https://doi.org/10.1109/wcnc57260.2024.10571121 Summary: Compares FSK with frequency-shift chirp modulation (FSCM, LoRa-like) in free-space and underwater VLC. FSCM beats FSK in band-limited, frequency-selective channels. Borrow: A good "beyond BFSK" future-work item (chirps), and an honest limitation to acknowledge for FSK.

D. Other modulations (for the comparison table) **17. "Designing A Dimmable OPPM-Based VLC System Under Channel Constraints,"** arXiv:1602.08765, 2016. *(abstract/intro read; authors not captured)* URL: https://arxiv.org/pdf/1602.08765 Summary: Overlapping PPM designed for dimming control and data rate under LED/channel bandwidth constraints. Cites the IEEE 802.15.7 flicker/dimming requirements. Borrow: PPM pros and cons for the comparison table (power efficiency, dimming).

18. S. D. Dissanayake, J. Armstrong, "Comparison of ACO-OFDM, DCO-OFDM and ADO-OFDM in IM/DD Systems," *Journal of Lightwave Technology*, vol. 31, no. 7, pp. 1063-1072, 2013. DOI 10.1109/JLT.2013.2241731 URL: https://opg.optica.org/jlt/abstract.cfm?uri=jlt-31-7-1063 · PDF mirror: https://optiwave.com/wp-content/uploads/2024/02/Comparison-of-ACO-OFDM-DCO-OFDM-and-ADO-OFDM-in-IM-DD-Systems.pdf Summary: The standard reference on making OFDM real and non-negative for LEDs. DCO-OFDM wastes optical power on DC bias, while ACO-OFDM wastes half the subcarriers. Borrow: One paragraph explaining why OFDM is the high-rate choice but out of scope for an MCU-based student link.

19. E. Monteiro, S. Hranilovic, "Design and Implementation of Color-Shift Keying for Visible Light Communications," *Journal of Lightwave Technology*, vol. 32, no. 10, pp. 2053-2060, 2014. DOI 10.1109/JLT.2014.2314358 URL: https://opg.optica.org/jlt/abstract.cfm?uri=jlt-32-10-2053 · thesis (open): https://macsphere.mcmaster.ca/handle/11375/13484 Summary: A CSK constellation design and hardware implementation with RGB LEDs. Total intensity stays constant, so CSK is inherently flicker-free. Borrow: The CSK row of your table, and the idea that "the colour equivalent of FSK" exists.

E. Low-cost Arduino/MCU VLC prototypes **20. (*) A. L. Rodrigues Gonçalves, D. Alves de Lima, Á. Maia**, "Visible Light Communication Applied to Low Cost Embedded Systems," *IEEE Latin America Transactions*, vol. 22, no. 1, pp. 15-21, 2024 (published online 21 Dec 2023).** (full page read) **URL: https://latamt.ieeer9.org/index.php/transactions/article/view/8232 · code: https://github.com/analuisarg/SistemaVLC Summary: A didactic two-Arduino-UNO VLC link with off-the-shelf parts, OOK + Manchester and a checksum, reaching 500 b/s at 40 cm. Hardware and code are published. Borrow:** Your OOK baseline.** Rebuild something like it on the same LED/photodiode so the BFSK-vs-OOK comparison is fair. Their low numbers (500 b/s, 40 cm) are an easy benchmark to beat and cite.

21. G. Dmitriev, K. Chumak, A. Bobrovsky, R. Andreev, O. Vorobyov, "Research of characteristics of Visible Light Communications system implemented on Arduino Uno platform," *2019 IEEE BlackSeaCom*, Sochi, June 2019. DOI 10.1109/BlackSeaCom.2019.8812781 *(abstract only)* URL: https://ieeexplore.ieee.org/document/8812781 Summary: Measures the boundary frequencies and link quality of Arduino-Uno optical TX/RX modules, and proposes circuit fixes to raise the optical clock and reduce ambient-light effects. Borrow: The test methodology (sweep the frequency until the link fails) and the ambient-light circuit ideas.

22. S. Schmid, G. Corbellini, S. Mangold, T. R. Gross, "An LED-to-LED Visible Light Communication System with Software-Based Synchronization," ETH Zurich / Disney Research Zurich (presentation slides, 2012). URL: https://www.bu.edu/smartlighting/files/2012/10/Schmid_.pdf Summary: Low-cost MCU LED-to-LED links (an LED used as both transmitter and receiver) for networked toys, with OOK and software synchronization, built around no-flicker and constant-brightness requirements. Borrow: Software clock-recovery ideas, and the "LED as receiver" trick as a stretch goal.

23. Q. Wang, D. Giustiniano, D. Puccinelli, "An open source research platform for embedded visible light networking," *IEEE Wireless Communications*, vol. 22, no. 2, pp. 94-100, Apr. 2015. DOI 10.1109/MWC.2015.7096291 URL: https://arxiv.org/abs/1410.0328 · project: http://www.openvlc.org/ (code: https://github.com/openvlc/OpenVLC/) Summary: OpenVLC is an open-source, software-defined VLC platform on a BeagleBone Black. Recent versions report about 400 kb/s at the transport layer and almost 20 m range (per openvlc.org). Borrow: Software-defined PHY/MAC structure and an open hardware front-end to compare your schematic with. Also a "related platforms" citation.

F. Ambient-light rejection and receiver front-ends **24. (*) F.-L. Chang, W.-W. Hu, D.-H. Lee, C.-T. Yu**, "Design and implementation of anti low-frequency noise in visible light communications," *2017 IEEE Int. Conf. on Applied System Innovation (ICASI)*, Sapporo, May 2017. DOI 10.1109/ICASI.2017.7988219 *(abstract only)* URL: https://ieeexplore.ieee.org/document/7988219 Summary: Receiver chain of TIA -> 4th-order Sallen-Key high-pass filter -> Schmitt trigger. It removes DC from sun/ambient light and suppresses low-frequency flicker from fluorescent lamps.**

Borrow: Your analog front-end almost exactly.** Replace their HPF with a band-pass centred between f0 and f1 and you have a BFSK front-end.

25. "Analog filters design in VLC analog front-end receiver for reducing indoor ambient light noise," *2016 IEEE Asia Pacific Conf. on Circuits and Systems (APCCAS)*. DOI 10.1109/APCCAS.2016.7804058 *(abstract only; authors not captured because the publisher page failed to load)* URL: https://doi.org/10.1109/apccas.2016.7804058 Summary: DC-offset removal (a 10 Hz HPF in automatic mode, or a pot plus differential amplifier in manual mode) plus a 100 Hz band-stop filter against lamp flicker, tested in simulation and in real time. Borrow: A 100 Hz notch (India mains is 50 Hz, so lamp flicker sits at 100 Hz) and a DC-removal stage before your tone filters.

26. "Ambient Light Rejection Using a Novel Average Voltage Tracking in Visible Light Communication System," *Applied Sciences*, vol. 7, no. 7, 670, 2017. DOI 10.3390/app7070670 *(abstract only; authors not captured)* URL: https://doi.org/10.3390/app7070670 Summary: An average-voltage-tracking threshold circuit that is insensitive to sunlight and indoor fluorescent light, aimed at low-SNR VLC receivers. Borrow: An adaptive threshold for your post-demodulator slicer (or the software equivalent on ESP32).

Additional front-end references seen (optional): S.-A. Avătămăniţei et al., logarithmic transimpedance VLC receiver, 50 m link in direct sun, *Sensors* 2020, 20, 909 (https://doi.org/10.3390/s20030909, *cited, not opened*); M. S. Islim et al., "The Impact of Solar Irradiance on Visible Light Communications," *JLT* 36, 2018 (https://doi.org/10.1109/JLT.2018.2813396, *cited, not opened*); W. Uemura, T. Hamano, "Noise Mitigation Methods for Digital Visible Light Communication," arXiv:2602.18187, 2026 (https://arxiv.org/abs/2602.18187v1: subtracts periodic AC-lamp noise and adds a second "noise-reference" photodiode, an ANC-style idea that is cheap to try).

G. VLC positioning 27. Y. Zhuang, L. Hua, L. Qi, J. Yang, P. Cao, Y. Cao, Y. Wu, J. Thompson, H. Haas, "A Survey of Positioning Systems Using Visible LED Lights," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 3, pp. 1963-1988, 2018. DOI 10.1109/COMST.2018.2806558 URL: https://doi.org/10.1109/COMST.2018.2806558 · accepted manuscript (open): https://uhra.herts.ac.uk/id/eprint/6968/1/Accepted Summary: A comprehensive VLP survey covering proximity, RSS, TDOA, AOA, image-sensor approaches and hybrids with inertial/RF sensors. Borrow: If you add positioning, FSK-ID proximity or RSS trilateration with 3-4 LEDs on different tone pairs (FDM) is the natural extension.

Also: **T.-H. Do, M. Yoo**, "An in-Depth Survey of Visible Light Communication Based Positioning Systems," *Sensors*, vol. 16, no. 5, 678, 2016. DOI 10.3390/s16050678 (open access). URL: https://www.mdpi.com/1424-8220/16/5/678. It is more tutorial-style and covers >100 papers, including FDM-based LED identification (good for a multi-tone design).

(*) The 6 closest to a BFSK student design

#	Source	Why it's close
9	Beguni et al., *Sensors* 2023	Actual BFSK VLC hardware, compared against OOK. Your template.
24	Chang et al., ICASI 2017	TIA -> active filter -> Schmitt receiver, exactly your analog chain.
20	Rodrigues Gonçalves et al., IEEE LatAm Trans 2024	Low-cost Arduino OOK-Manchester baseline with open code.
13	Rajagopal, Lazik, Rowe, IPSN 2014	Low-kHz FSK tones from LEDs, tone-spacing and flicker numbers, positioning extension.
11	Mougoue Yamga et al., ICAST 2018	Low-complexity (square-wave) FSK theory for non-negative LED signals.
10	Azim et al., IEEE TCCN 2019	Rigorous DC-FSK/U-FSK theory and BER expressions to compare against.

(Runner-up: #12 Kang et al. ICPADS 2024, for a recent low-power-LED M-FSK range benchmark.)

2. Modulation comparison for VLC (IM/DD with LEDs)

Ratings are qualitative and relative, synthesised from sources 2, 6, 9, 10, 17, 18, 19 and the Photonics 2023 survey (https://doi.org/10.3390/photonics10090993). "Student MCU build" means an Arduino/ESP32-class system without an FPGA or SDR.

Scheme	Data rate (spectral efficiency)	Noise / ambient-light immunity	Flicker / dimming	Complexity	Cost	Typical student-build reality
OOK (NRZ)	Medium (1 bit/symbol). Rate limited by LED bandwidth.	**Poor.** Energy sits near DC to low kHz, overlapping sunlight DC and 100 Hz to kHz lamp flicker. Needs an adaptive threshold.	Long runs of 0s/1s cause visible flicker. Dimming only via compensation symbols.	Lowest	Lowest	kb/s on Arduino (e.g. 500 b/s at 40 cm in #20)
Manchester OOK	$\frac{1}{2}$ of NRZ (2 chips/bit)	Better than NRZ: no DC, so a high-pass can remove ambient DC. Still overlaps low-frequency lamp noise at low rates.	Good: 50% duty, flicker-free above ~200 Hz chip rate. Used in 802.15.7 PHY I.	Low (clock recovery needed)	Lowest	The fair baseline to compare BFSK against (#9 did exactly this).
BFSK (DC-FSK / square-tone FSK)	**Low** (1 bit/symbol, but each bit needs several carrier cycles, so the bit rate is well below the tone frequency)	**Good.** Energy sits at two chosen tones, so you can place them above lamp interference and use narrow band-pass filters. ~20-25% better noise tolerance than OOK-Manchester measured in #9.	**Excellent.** Constant 50% average intensity whatever the data, so no data-dependent flicker if tones are > a few kHz.	Medium: two filters + PLL/tone decoder, or DSP (Goertzel)	Low (XR2211 or ESP32 only)	Realistic: 1-10 kb/s over a few metres with ESP32 or XR2211

Scheme	Data rate (spectral efficiency)	Noise / ambient-light immunity	Flicker / dimming	Complexity	Cost	Typical student-build reality
PPM / VPPM	Low-medium ($\log_2$L bits per L slots). Bandwidth-hungry.	Good power efficiency (high peak, low average), but needs tight slot timing and is sensitive to multipath at high rates.	VPPM in 802.15.7 gives dimming via pulse width.	Medium (slot sync)	Low	Feasible on MCU timers, but synchronization is the hard part.
OFDM (DCO / ACO-OFDM)	**Highest** (QAM on many subcarriers; Gb/s demos)	Good against frequency-selective channels, and can notch interfered subcarriers. DCO wastes power on bias.	Fine (high-frequency, DC-biased).	**High**: IFFT/FFT, DAC/ADC at MHz, linear LED driver, PAPR issues	High (FPGA/SDR, fast DAC/ADC)	Out of scope for MCU. Simulation-only for most students.
CSK	Medium-high ($\log_2$M bits/symbol with RGB LEDs)	Moderate. Colour crosstalk and ambient colour shifts need calibration.	**Excellent.** Total intensity constant, so inherently flicker-free.	High (RGB driver, 3-channel colour receiver, calibration)	Medium	Possible with RGB LED + colour sensor at low rates, but calibration-heavy.

Honest pros and limits of BFSK for this project **Pros** - **Ambient-light robustness by design.** Sunlight is DC, and incandescent lamps (on India's 50 Hz mains) produce 100 Hz with harmonics to ~2 kHz. Fluorescent/CFL interference extends to ~40 kHz with low-level components to ~1 MHz (per #9). Two tones placed in a clean band, behind a band-pass filter, reject most of this. A high-pass alone does not do that for OOK at low bit rates. - **Flicker-free regardless of data.** The LED is always toggling at f0 or f1 with 50% duty, so average brightness never changes with the bit pattern. - **Non-coherent, asynchronous detection** (PLL/tone decoder or Goertzel energy comparison). No carrier phase recovery is needed, which suits free-running MCUs (#9 makes this argument against PSK). - **Cheap, well-documented demodulators** (XR2211 up to 300 kHz, LM567 up to 500 kHz centre frequency), plus an all-digital path (Goertzel) that needs no extra ICs. - **Natural multi-user/positioning extension:** give each LED its own tone pair (FDM), as in #13.

Limits - **Low data rate / poor spectral efficiency.** You need several carrier cycles per bit, and the LED (especially a phosphor white LED) and photodiode/TIA bandwidth caps the tone frequency. #9 reached 10 kb/s with ~1 MHz tones and a dedicated demod IC. Expect 1-10 kb/s on ESP32. - **More hardware/tuning than OOK.** Two filters or a PLL, careful component tolerances (XR2211/LM567 centre frequency drifts with R/C), and calibration. #9 explicitly reports adjustment difficulty and jitter. - **Wasted optical power:** the DC bias / 50% duty carries no information (the same DC-FSK issue analysed in #10). - **PLL lock time and capture range** limit how fast you can switch tones (bit rate vs loop filter). With Goertzel, the block length N sets both frequency resolution (fs/N) and minimum bit duration. - **Band-limited channels favour other schemes** (#16 shows chirp modulation beating FSK there). BFSK is the "robust and simple" choice, not the "fast" one.

3. Research gap a student project can address

What the literature already has: - High-rate VLC (OFDM, CSK) is mostly simulation or lab-grade hardware (FPGA/SDR) (#18, #19; noted in #9). - The one solid BFSK *hardware* comparison (#9) uses ~1 MHz tones, a dedicated analog FSK demod IC (KA3361), a 600 MHz ARM board, a high-power red LED with an optical filter, and halogen noise in a European lab. - Low-cost Arduino VLC papers (#20, #21) are almost all OOK/Manchester with no systematic ambient-light study. - FSK theory papers (#10, #11) are simulation-only.

Gap (pick one as the headline, the rest as sub-objectives):

1. **Analog PLL vs digital Goertzel BFSK demodulation, same link, same noise.** No source found compares an XR2211/LM567 hardware demod head-to-head with a Goertzel detector on a low-cost MCU (ESP32) for VLC, measuring BER vs distance and vs ambient illuminance (lux). This is cheap, novel enough for a conference paper, and fits your skills.
2. **BFSK vs Manchester-OOK on identical low-cost hardware under *real Indian indoor lighting*:** 50 Hz mains (100 Hz flicker), CFL/LED-tube drivers, and tropical daylight through windows. Record the interference spectrum with the ESP32 ADC (or a sound card / oscilloscope FFT) and report BER vs lux for both schemes. #9 did this with halogen only.
3. **Interference-aware (adaptive) tone selection:** the receiver measures the noise spectrum (a Goertzel bank at candidate frequencies) during idle time and the link hops to the cleanest f0/f1 pair. This is a simple "cognitive VLC" idea that #10's journal (cognitive comms) hints at, with no low-cost hardware demo found.
4. **White phosphor LED + DC-FSK dimming:** show that BFSK keeps illumination constant while dimming (vary the DC bias or duty) and measure the BER cost. This is relevant to real room lighting, whereas #9 used a red LED.
5. **Stretch goal: FSK-ID beacons for indoor positioning** (#13, #27): 3 LEDs on 3 tone pairs, with RSS-based position estimation on the ESP32.

Suggested title: "Low-Cost ESP32-Based BFSK Visible Light Communication Link with Goertzel Demodulation: Experimental Comparison with PLL Demodulation and OOK Under Indoor Ambient Light."

Suggested measurable outcomes: BER (target $< 10^{-4}$ at the chosen rate) vs distance (0.5-5 m), BER vs ambient lux (dark, room light, window daylight), max bit rate for each demodulator, power consumption, and BOM cost (in Rs.).

Design starting points (to verify on the bench): - Keep tones well above the ~2 kHz flicker/harmonic zone. Either stay below the main CFL band (e.g. ~5-20 kHz, easy for ESP32 ADC sampling) or go above it ($\geq$ 50 kHz, needs a faster TIA and the ESP32 ADC continuous/I2S mode near its limit, or the XR2211, max 300 kHz). - Space f0 and f1 at $\geq$ the bit rate for non-coherent orthogonality (#9 uses Rb = 1/Ts spacing). - With Goertzel, choose fs and block length N so that f0 and f1 land on bin centres (k = N·f/fs) and N/fs $\leq$ bit time. - The exact ESP32 ADC maximum sample rate depends on the chip variant and mode, so confirm it in the Espressif ADC continuous-mode docs (link in §4) before fixing tone frequencies.

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4. Tutorials and learning resources (15 verified links)

Receiver analog front-end (photodiode + TIA) 1. TI, *Transimpedance Amplifier Circuit* (Analog Engineer's Circuit SBOA268B, P. Semig): step-by-step Rf/Cf selection and a bandwidth formula. https://www.ti.com/lit/an/sboa268b/sboa268b.pdf 2. TI, *1 MHz, Single-Supply, Photodiode Amplifier Reference Design* (TIDU535): a single 5 V supply TIA with bias to avoid rail saturation, matching your hardware. https://www.ti.com/lit/ug/tidu535/tidu535.pdf 3. TI, *Transimpedance Amplifiers (TIA): Choosing the Best Amplifier for the Job* (SNOA942A): noise vs bandwidth trade-offs and op-amp selection. https://www.ti.com/jp/lit/an/snoa942a/snoa942a.pdf 4. Analog Devices, *Stabilize Your Transimpedance Amplifier*: how to pick the feedback capacitor for phase margin (prevents a ringing/oscillating TIA). https://www.analog.com/en/resources/technical-articles/stabilize-transimpedance-amplifier-circuit-design.html 5. Analog Devices, *Optimizing Precision Photodiode Sensor Circuit Design* (L. Orozco): photovoltaic vs photoconductive mode and noise analysis. https://www.analog.com/en/resources/technical-articles/optimizing-precision-photodiode-sensor-circuit-design.html

Analog BFSK demodulators 6. MaxLinear/Exar **XR-2211A FSK Demodulator/Tone Decoder datasheet**: PLL FSK demod, 0.01 Hz-300 kHz, 4.5-20 V, with design equations for R0/C0 (centre frequency), R1 (tracking

range), C1 (loop filter) and a carrier-detect connection. https://www.maxlinear.com/ds/xr2211av104.pdf 7. TI **LM567x Tone Decoder datasheet** (Rev. F): f0 ≈ 1/(1.1·R1·C1), up to 500 kHz, bandwidth 0-14%. Use two LM567s (one per tone) as a simple BFSK detector. https://www.ti.com/lit/ds/symlink/lm567c.pdf

Digital demodulation: Goertzel 8. K. Banks, "The Goertzel Algorithm," *Embedded.com*: the classic embedded-programmer explanation (block size N, sampling-rate choice, the optimized Goertzel). https://www.embedded.com/the-goertzel-algorithm/ 9. DSP Workshop, "Goertzel on Hardware": C struct and per-sample update code, real-only power, no buffer needed. Drops straight into ESP32 firmware. https://dsp-workshop.nl/topics/goertzel/embedded.html 10. TI Application Report SPRA066, *Modified Goertzel Algorithm in DTMF Detection*: a formal derivation plus a detection-logic example (tone presence thresholds). https://www.ti.com/lit/an/spra066/spra066.pdf

ESP32 peripherals 11. Espressif ESP-IDF, **LED Control (LEDC)**: generate the f0/f1 square-wave tones on a GPIO and change frequency per bit. https://docs.espressif.com/projects/esp-idf/en/stable/esp32/api-reference/peripherals/ledc.html (Arduino-core version: https://docs.espressif.com/projects/arduino-esp32/en/latest/api/ledc.html) 12. Espressif ESP-IDF, **ADC Continuous Mode Driver**: DMA-based high-rate sampling of the TIA output for Goertzel. https://docs.espressif.com/projects/esp-idf/en/stable/esp32/api-reference/peripherals/adc/adc_continuous.html (Related: I2S driver page, which notes the original ESP32 can route I2S0 to the internal ADC/DAC: https://docs.espressif.com/projects/esp-idf/en/stable/esp32/api-reference/peripherals/i2s.html)

Courses 13. NPTEL, **Optical Wireless Communications for Beyond 5G Networks and IoT** (Prof. Anand Srivastava, IIIT Delhi, 12 weeks): LEDs, photodetectors, SNR, indoor channel models and modulation. Prerequisite: Digital Communication. https://onlinecourses.nptel.ac.in/noc23_ee61/preview · lecture list: http://www.digimat.in/nptel/courses/video/108106190/108106190.html 14. NPTEL, **Principles of Digital Communication** (Prof. Abhishek Dixit, IIT Delhi): FSK/non-coherent detection and BER theory for your analysis chapter. https://nptel.ac.in/courses/108102120

Videos and builds 15. Harald Haas, **"Wireless data from every light bulb"** (TEDGlobal 2011), the talk that coined "Li-Fi" (good intro for your viva). https://www.ted.com/talks/harald_haas_wireless_data_from_every_light_bulb · follow-up "Forget Wi-Fi. Meet the new Li-Fi Internet" (2015): https://www.ted.com/talks/harald_haas_forget_wi_fi_meet_the · YouTube "LiFi: What is it?": https://www.youtube.com/watch?v=FbDohcbuhu0

Build write-ups (OOK-style, useful for mechanics and baselines): - Instructables, *Visible Light Communication System* (Teensy 4.1 + Arduino, Fresnel lens for range): https://www.instructables.com/Visible-Light-Communication-System - Instructables, *Li-Fi Audio Transmission Through Light* (analog intensity modulation; an easy first demo): https://www.instructables.com/LiFi-Experiment-Audio-Transmission-Through-Light - Hackaday, *Hackaday Explains: Li-Fi & Visible Light Communications* (2015): https://hackaday.com/2015/12/11/hackaday-explains-li-fi-visible-light-communications/ - Hackaday.io, *Power LED LiFi Communication for IoT Smarthomes* (custom TI LM3409 fast-switching power-LED driver, clean switching to ~120 kHz, which shows how to drive a bright LED fast): https://hackaday.io/project/26181-power-led-lifi-communication-for-iot-smarthomes

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5. Commercial Li-Fi products and companies

Company	Product / link	One line
pureLiFi (UK; co-founded by Prof. Harald Haas)	link · Kitefin XE: link	Kitefin XE is a room-filling LiFi access point (>80 m^2) with USB clients for Android/Windows/Linux, aligned to IEEE 802.11bb.
Signify (Philips) - Trulifi (Netherlands)	link · India page: link · datasheets: link	Trulifi 6002 (meeting rooms, ~220/160 Mb/s down/up, roaming), 6013/6014 point-to-point (industrial), 6004 (first LiFi system with a FIPS 140-3 validated crypto engine, June 2025: link). Note: the Trulifi systems listed are infrared LiFi.

Company	Product / link	One line
Oledcomm (France; started as a university lab in 2005)	LiFiMAX: link · product range: link	LiFiMAX access points/antennas (~150 Mb/s), USB receivers, a LiFi tablet, a point-to-point unit (up to 5 Gb/s at 5 m per their page), and the SOLERIS defence range.
Velmenni (India-founded, New Delhi, 2012/2014; now also Estonia/US)	link · About: link · Dongle/APs: link	Describes itself as the first Indian light-communication company. Early "Jugnu" Li-Fi smart bulb; today mostly FSO/Li-Fi wireless backhaul (raised Rs.30 crore pre-Series A in Mar 2026, per BusinessLine: link). Good Indian industry contact for internships.
Fraunhofer IPMS (Germany, research institute with products)	link	Li-Fi HotSpot and GigaDock modules for industrial low-latency links, plus Power-over-Light (data + power up to 10 Mb/s, 1 m).
LiFi Group (Netherlands, reseller/integrator)	link	Signify partner that maintains a useful directory of Li-Fi companies (handy for your "industry landscape" slide).

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6. Where to publish a student paper

Dates checked 31 Oct 2026. Verify deadlines on the official sites before planning.

Venue	Type / publisher	Link	Status / notes
NCC (National Conference on Communications)	India's flagship comms conference (IITs + IISc JTG), IEEE-format, 6 pages, double-blind	NCC 2027 CFP: link · archive: link	NCC 2027: IISc Bengaluru, 17-20 Feb 2027. Paper deadline was 16 Oct 2026 (passed, per an organiser's LinkedIn post). **Target NCC 2028** (deadline usually around Sept-Oct). It has an "RF and Optical" track. Most prestigious India option.
IEEE INDICON	IEEE India Council flagship, proceedings on IEEE Xplore	link · link	INDICON 2026: Sri Sairam Engineering College, **Chennai**, 18-20 Dec 2026. Submissions closed (deadline was 31 Aug 2026). Target **INDICON 2027** (deadline typically mid-year). Student fee ~Rs.9,000 (2026 rates).

Venue	Type / publisher	Link	Status / notes
IEEE CONECCT	IEEE Bangalore Section flagship, 6-8 pages, includes a Communication Systems focus area and poster/demo tracks	link	**CONECCT 2027: 29-31 July 2027, Bengaluru** (per the site home page). 2026's paper deadline was in Feb, so expect ~Feb 2027. **Best timing fit** for a project finishing in spring 2027. Poster/demo tracks suit hardware prototypes.
ICCCNT	IEEE-sponsored, large multi-track (computing/communication/networking)	link · EasyChair CFP: link	17th ICCCNT 2026 was listed for IIT Delhi (28 Jun-3 Jul 2026), but the EasyChair CFP lists UNT, Texas, 19-21 Dec 2026. **The sources conflict**, so confirm on the official site. High acceptance, student-friendly. (!) Do **not** confuse it with the WASET "ICCCNT" listings at waset.org (widely regarded as predatory) or with "icICCNT" (Paris).
Springer LNEE (Lecture Notes in Electrical Engineering)	Book series publishing many Indian conference proceedings	link	Not a conference itself: you submit to a conference whose proceedings go to LNEE. Scopus-indexed (SJR Q4 per link). Check that the conference is genuine before paying fees.
MDPI Sensors / Photonics (journals, open access)	Peer-reviewed, Scopus/WoS indexed, with APCs (article processing charges)	Sensors: link · the BFSK paper #9 appeared in its *Automotive VLC* special issue	The natural home for an extended version with a strong hardware/experimental component. APCs are significant, so check for waivers or institutional support.
IEEE Latin America Transactions	IEEE journal, accepts didactic/low-cost embedded work (#20 was published there)	link	Shows that "low-cost educational VLC" papers are publishable in IEEE journals.

Recommended path: build and measure through Feb 2027, then submit to **IEEE CONECCT 2027** (Bengaluru, July 2027) or the next ICCCNT. Aim for **NCC 2028 / INDICON 2027** if results are strong. Consider an MDPI Sensors extension later.

Gaps and caveats in this pack

- Some IEEE papers were read only at abstract/metadata level (paywalled). Full citations for #15, #16, #17, #25 and #26 lack author names because the publisher pages did not show them during this session. Look

them up via the DOI before citing.
- Kahn & Barry (#8), Avătămăniţei 2020 and Islim 2018 were seen only as DOIs in reference lists.
- The IEEE 802.15.7 flicker threshold (commonly quoted as a 5 ms maximum flicker period, ≈200 Hz) was **not** verified from the standard text in this session, so check it in source #1/#2 before quoting a number.
- The ESP32 ADC maximum sampling rate was not extracted, so check the ADC continuous-mode docs for your exact chip (ESP32 vs ESP32-S3).
- Conference dates and fees change. Re-check the official sites before planning around them.

Part C: Hardware Design

Li-Fi BFSK link: hardware design pack

Student visible-light link using binary FSK. Solo build, India, under Rs 3,000. Target: 1 to 10 kbps over 0.5 to 3 m indoors, with rejection of ambient light. Lab gear assumed: oscilloscope, function generator, bench or USB 5 V supply, multimeter.

Files in `/workspace/work/lifi/hardware/`:

File	What it is
`hardware.md`	This document
`rx_frontend.cir`	RX analog front end netlist for LTspice and ngspice: BPW34, TIA, DC servo, gain stage, MFB band-pass (branch A), Sallen-Key anti-alias filter (branch B)
`tx_driver.cir`	TX LED driver netlist: ESP32 GPIO, IRLZ44N, 1 W LED. Also contains an AO3400 model for tiers B and C
`rx_ac_response.png`	ngspice AC response of every RX stage
`rx_noise.png`	ngspice input-referred noise of the RX circuit
`tx_transient.png`	ngspice transient of the TX driver at 15 kHz
`sim/rx_ac.sp`, `sim/tx_tran.sp`	ngspice batch runners (use `.include ../*.cir`)
`sim/make_plots.py`, `sim/link_budget.py`, `sim/link_budget.md`	Plot and link-budget scripts and their output
`ngspice.sh`	Sandbox-only wrapper for a user-space ngspice 44.2. You don't need it on your PC

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0. Design decisions at a glance

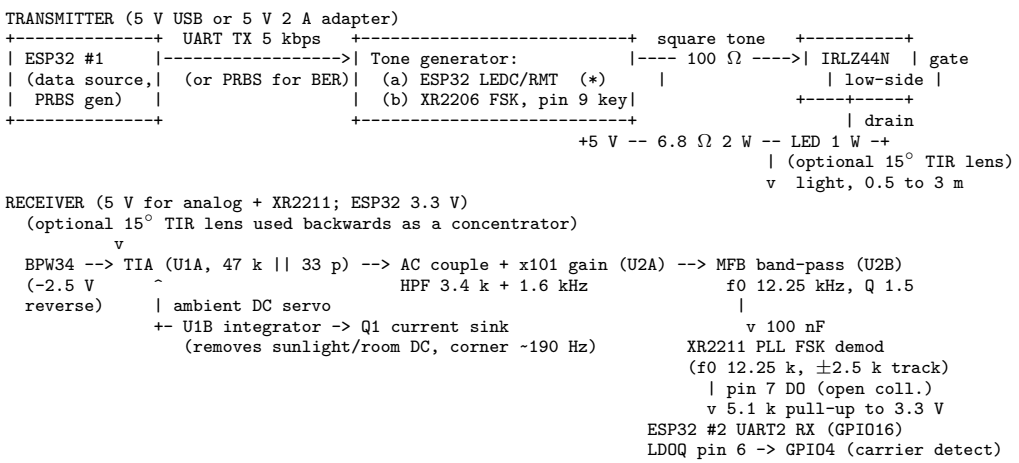
Item	Choice	Why
Tone plan, main	**f1 (logic 1, mark, idle) = 10 kHz; f0 (logic 0, space) = 15 kHz**	Far above the 100 Hz mains flicker and its strong low harmonics. Above 3 kHz, so no visible or stroboscopic flicker. Inside the MCP6022 bandwidth at ×101 gain. Both demodulators can use it, which gives a fair PLL vs Goertzel comparison.
Bit rates on the main plan	2.5 kbps (safe) and **5 kbps (comparison point)**	$\Delta f = 5$ kHz = Rb at 5 kbps, so the tones are orthogonal for non-coherent FSK. That is ≥ 2 cycles of the low tone per bit, so the XR2211 PLL can follow.
Tone plan, 10 kbps (tier B)	20 / 30 kHz, Goertzel on ESP32	Δf = Rb = 10 kHz. With fs = 200 kS/s, N = 20 samples per bit, bins k = 2 and 3 are integers.

Item	Choice	Why
Bring-up plan	1200 / 2200 Hz at 1200 bps (Bell-202)	This is the XR2211 datasheet's own worked example, so it verifies the chip straight from the function generator.
LED	1 W white star LED, switched on/off (square tone), about 0.27 A peak	Cheap, bright, no driver IC needed. The average stays constant, so illumination looks steady.
Switch	IRLZ44N (TO-220, logic level) for tier A; AO3400 (SOT-23) for B and C	IRLZ44N fits a breadboard. The AO3400 has 1/3 the gate charge, so its edges are faster.
Photodiode	BPW34 PIN, 7.5 mm^2, reverse-biased 2.5 V	Rs 59 to 92 in India. 25 pF at 3 V, 100 ns rise time.
Op-amp	**MCP6022** (dual, 10 MHz, 1 pA bias, rail-to-rail, 2.5 to 5.5 V, DIP-8)	It works on 5 V and 3.3 V, so its output can't damage the ESP32 ADC. Trade-offs are in section 2.3.6.
Ambient rejection	(1) DC servo sinks ambient photocurrent at the TIA input (headroom up to about 400 μA). (2) Two AC-coupled high-pass corners. (3) Band-pass or Goertzel bins.	Simulated: 100 Hz flicker sits 105 dB below the in-band gain at the XR2211 input (section 3)

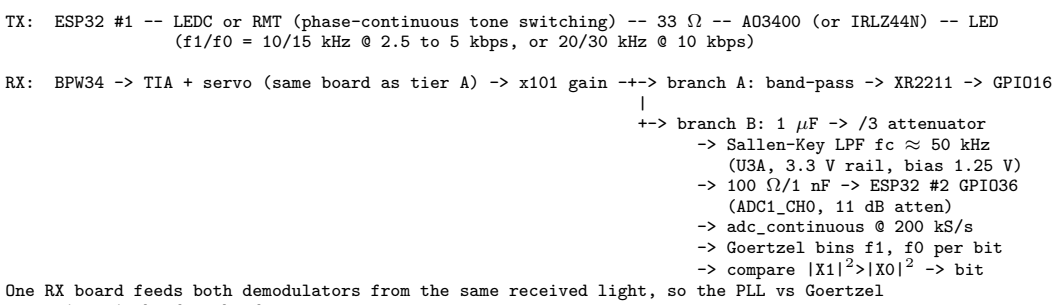
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1. System block diagrams

1A. Tier A: analog/IC baseline (XR2211 PLL demodulator)



1B. Tier B: MCU-centric (ESP32 LEDC tone generation + ADC/I2S sampling + Goertzel)



1C. Tier C: stretch, higher rate (40 to 500 kbps)

```
TX: ESP32 RMT/SPI bitstream - 22 Ω - AO3400 - LED (5 mm high-brightness RED or BLUE, 30 to 50 mA,
                                                   or 1 W red; no phosphor) with an RC "speed-up"
                                                   network: R_series || 1 nF
    Modulation: BFSK 100/150 kHz @ 50 kbps (edge-period demod) or Manchester OOK @ 250 to 500 kbps

RX: BPW34 (5 V reverse bias, Cd ≈ 20 pF) -> TIA Rf 10 k || 4.7 pF (MCP6022, BW ≈ 2 MHz)
    -> AC couple (1 nF/10 k, 16 kHz HPF) -> ×5 gain -> LM311 comparator with 20 mV hysteresis
    -> ESP32: RMT-RX or PCNT measures each period (FSK) / UART at 250 to 500 kbaud (Manchester OOK)
    (optional: blue gel filter on the PD to block the slow yellow phosphor light of a white LED)
```

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2. Schematics, component by component, with calculations

2.1 Power and references

Ref	Part	Connection	Notes
PS1	5 V 2 A adapter or USB charger	TX: +5 V rail for the LED and the ESP32 VIN	LED peak 0.27 A plus ESP32 about 0.1 A
RX 5 V	from the RX ESP32 5 V/VIN pin (USB)	MCP6022 U1/U2, XR2211 VCC	Add 100 nF at every IC VDD pin and 10 μF bulk per board
RX 3.3 V	ESP32 3V3 pin	U3 (branch B) only	Keeps the ADC input at or below 3.3 V
VREF 2.5 V	10 k / 10 k divider from 5 V + 10 μF + 100 nF	non-inverting inputs of U1A, U1B, U2B; RHP, CG, R2BP returns	Its AC impedance must be low. If you see hum on VREF, buffer it with a spare op-amp half.
VB2 1.25 V	22 k (to 3V3) / 13 k (to GND) + 10 μF	branch B bias	$3.3 \cdot 13/35 = 1.23$ V
GND	single star point at the RX board		Keep the TX LED current loop away from RX ground. Use separate boards.

2.2 Transmitter

2.2.1 LED driver (tier A and B), simulated in `tx_driver.cir`

Ref	Value / part	Pins / nodes
U_TX	ESP32 DevKit (38-pin)	**GPIO25** = tone output. Avoid the strapping pins 0, 2, 12, 15.
RG	100 Ω, $\frac{1}{4}$ W	GPIO25 -> MOSFET gate
RGS	100 kΩ	gate -> GND. Keeps the LED off while the ESP32 boots or floats.
Q_TX	IRLZ44N (TO-220: pin 1 G, pin 2 D, pin 3 S, tab = D)	S -> GND; D -> LED cathode
D_LED	1 W white LED on a 20 mm star	anode -> R_LED; cathode -> drain
R_LED	6.8 Ω, 2 W metal-oxide	+5 V -> LED anode
C_bulk	470 μF/10 V + 100 nF across +5 V/GND, right at the LED	supplies the 0.27 A current pulses

Calculations

- LED current: I = (5 - V_F - V_DS) / R = (5 - 3.2 - 0.01) / 6.8 ≈ **0.26 A peak** (simulation: 0.267 A). At 50 % duty the average is about 0.13 A, so the LED is far below its 350 mA rating and runs cool on the star.

Because V_F of cheap LEDs spreads from 2.9 to 3.5 V, **measure the drop across R_LED** and adjust R (5.6 to 8.2 Ω) for 0.25 to 0.3 A.

- R_LED dissipation: $I^2R = 0.26^2 \cdot 6.8 = 0.46$ W peak, 0.23 W average, so a 2 W part has margin.
- MOSFET: R_DS(on) ≤ 35 mΩ at V_GS 4.5 V (Infineon). At 3.3 V it is somewhat higher, about 50 mΩ typical. Conduction loss = $0.26^2 \cdot 0.05 \cdot 0.5 \approx 2$ mW. No heatsink needed.
- Gate resistor: the ESP32 pin's default drive is 20 mA, with an absolute maximum of 40 mA. RG ≥ 3.3 V / 25 mA - R_pin(≈30 Ω) ≈ 100 Ω. IRLZ44N gate charge is about 32 nC at 4.5 V (about 20 nC at 3.3 V), so the gate rise ≈ Q / I_avg ≈ 20 nC / 13 mA ≈ 1.5 µs. **Simulated:** gate 10 to 90 % in 0.92 µs, drain fall 82 ns, turn-on delay 0.21 µs. At 15 kHz (33 µs half-period) that is under 3 % of the period, which is fine. At 20/30 kHz or in tier C, use the **AO3400** (Q_g ≈ 6 nC, C_iss 630 pF, V_th 0.65 to 1.45 V) with RG = 22 to 33 Ω. The edges then become sub-200 ns.
- LED bandwidth: phosphor-converted white LEDs have about 1 to 3 MHz total modulation bandwidth. The yellow phosphor (~0.5 to 1 MHz) is the slow part; the blue die reaches 10 to 20 MHz. At 10 to 30 kHz the LED is effectively instantaneous. It only matters in tier C, which uses a blue filter at the RX or a red/blue LED.
- Optional 15° TIR lens: on axis, the intensity gain ≈ η(m'+1)/(m+1). With m' = -ln2 / ln cos 7.5° ≈ 81 and $\eta \approx 0.8$, that is about **30× (+15 dB optical, +30 dB electrical)**. The beam at 3 m is only about 0.8 m wide, so the lens is for range tests, not for coverage.

2.2.2 Tone generation, option (a), recommended: ESP32 LEDC/RMT

- LEDC at 50 % duty: `ledcAttach(25, 10000, 8)`. Switching frequency with `ledcChangeFrequency` causes phase glitches and takes tens of µs. So for 5 to 10 kbps, generate the waveform with **RMT** (each half-period is one RMT item: 50 µs / 33.3 µs) or with a 1 MHz timer ISR. Both give phase-continuous CPFSK and exact frequencies. The details belong to the firmware folder.
- Mapping: idle = mark = logic 1 = **10 kHz**; logic 0 = **15 kHz**. With this, the XR2211 data output (high for the lower frequency) directly reproduces UART polarity.

2.2.3 Tone generation, option (b): XR2206 analog FSK The datasheet pinout is listed here; check it against your copy before you power up.

XR2206 pin	Connect
4 V_CC	+12 V. The XR2206 needs 10 to 26 V single supply, so it needs the 12 V adapter.
12 GND	GND
5, 6 TC1/TC2	C_T = 10 nF film between them
7 TR1 (selected when pin 9 is high or open)	R = 8.2 k + 5 k trimpot to GND -> f = 1/(R·C) = 10 kHz needs R = 10.0 k
8 TR2 (selected when pin 9 < 1 V)	R = 5.6 k + 2 k trimpot to GND -> 15 kHz needs R = 6.67 k
9 FSK input	MCU UART TX (3.3 V logic is OK: > 2 V selects pin 7)
10 BIAS	1 µF to GND
11 SYNC (square out, open collector)	1 kΩ pull-up to +5 V -> 47 Ω -> IRLZ44N gate. Don't use a 10 k pull-up: with the 1.7 nF gate the edges become 17 µs.

2.3 Receiver analog front end (simulated in `rx_frontend.cir`) MCP6022 DIP-8 pinout: 1 OUTA, 2 -INA, 3 +INA, 4 VSS (GND), 5 +INB, 6 -INB, 7 OUTB, 8 VDD.

2.3.1 Photodiode and TIA (U1A)

Ref	Value	Connection
PD1	BPW34	**Cathode -> +5 V**, anode -> U1 pin 2 (TIN). Find the cathode with a multimeter diode test: the PD reads about 0.4 to 0.6 V when the red probe is on the anode.
U1A	MCP6022 half A	pin 3 = VREF 2.5 V, pin 2 = TIN, pin 1 = TOUT
RF	47 kΩ, 1 % metal film	pin 1 <-> pin 2
CF	33 pF C0G	pin 1 <-> pin 2

The photodiode sits at 5 - 2.5 = 2.5 V reverse bias, so C_d ≈ 25 to 30 pF. Light -> current into TIN -> V_TOUT = 2.5 V - I·RF (the output falls with light).

- **Gain choice.** RF sets both transimpedance and headroom. Without the servo, the output saturates at I_amb ≈ 2.4 V / 47 k ≈ 50 μA. That is roughly a brightly lit desk near a window: indoor 500 lux gives about 3 to 4 μA, indirect daylight 1 to 5 klux gives 10 to 50 μA. At 3 m the signal (≈ 20 nA peak) gives only about 1 mV at TOUT, so the real gain is placed after the AC coupling. A larger RF gains almost no SNR, because the shot noise of the ambient light already dominates. At 10 μA ambient: $\sqrt{(2qI)}$ = 1.8 pA/$\sqrt{\text{Hz}}$, versus 4kT/RF at 47 k = 0.59 pA/$\sqrt{\text{Hz}}$.
- **Stability (minimum CF).** C_in = C_d (30 p) + op-amp C_CM + C_DM (≈ 9 p) + Q1 C_cb (≈ 4 p) + stray (3 p) ≈ 45 pF. CF_min ≈ $\sqrt{(\text{C_in} / (2\pi \cdot \text{RF} \cdot \text{GBW}))}$ = $\sqrt{(45\text{ p} / (2\pi \cdot 47\text{ k} \cdot 10\text{ MHz}))}$ ≈ **3.9 pF** for about 45° phase margin. The chosen 33 pF gives far more margin and deliberately limits bandwidth: f_-3dB = $1/(2\pi \cdot 47\text{ k} \cdot 33\text{ p})$ = **103 kHz** (simulated 103.5 kHz). That passes 30 kHz tones and cuts noise.
- **Responsivity.** The BPW34 peaks at about 0.6 A/W near 900 nm. Over a white-LED spectrum (450 nm blue spike plus 550 to 650 nm phosphor) the effective value is about **0.3 A/W**, which is the number used in the link budget. The OPT101 is specified at 0.45 A/W at 650 nm.

2.3.2 Ambient-light DC servo (U1B + Q1)

Ref	Value	Connection
RI	100 kΩ	TOUT -> U1 pin 6
CI	39 nF film	U1 pin 7 <-> pin 6
U1B	MCP6022 half B	pin 5 = VREF, pin 7 = IOUT
RB	1 kΩ	IOUT -> Q1 base
Q1	BC547B (TO-92: C-B-E, flat side facing you)	collector -> TIN, emitter -> RE
RE	10 kΩ	Q1 emitter -> GND

How it works: when ambient light pulls TOUT below VREF, the integrator output rises and Q1 sinks I_C = (V_B - 0.65)/RE from the TIA input. That cancels the DC photocurrent, so TOUT returns to 2.5 V.

- Loop gain L(s) = RF / (RE·s·RI·CI), giving a first-order high-pass at the TIA output with f_c = RF / (2π·RE·RI·CI) = 47 k / (2π·10 k·100 k·39 n) ≈ **190 Hz**. Simulated: -3 dB at 153 Hz, -5 dB at 100 Hz.
- Capacity: I_max ≈ (4.9 - 0.65) / 10 k ≈ **0.4 mA**, which covers bright indirect daylight. Direct sunlight on a bare BPW34 (about 1 to 2 mA) still saturates it: shade the PD with a 2 to 3 cm black tube.
- Built-in ambient meter: **V(RE) / 10 kΩ = ambient photocurrent** (10 μA -> 100 mV). Log it on the scope or with a spare ESP32 ADC pin for every experiment.
- Noise cost: Q1 adds shot noise equal to the ambient's, a $\sqrt{2}$ penalty (about 1.5 dB SNR). That buys a 10× wider ambient range. For dim-room tests you can pull Q1 out.

2.3.3 AC-coupled gain stage (U2A), ×101

Ref	Value	Connection
CC1	4.7 nF film	TOUT -> U2 pin 3 (HIN)
RHP	10 kΩ	HIN -> VREF. First HPF corner: $1/(2\pi \cdot 10\text{ k} \cdot 4.7\text{ n})$ = **3.39 kHz**
RG2	100 kΩ	U2 pin 1 (GOUT) <-> pin 2
RG1 + CG	1 kΩ in series with 100 nF	pin 2 -> VREF. Gain = 1 + 100 k/1 k = 101 (40 dB) above $1/(2\pi \cdot 1\text{ k} \cdot 100\text{ n})$ = **1.59 kHz**; unity gain at DC, so no offset amplification

Closed-loop bandwidth ≈ GBW/101 ≈ 99 kHz (simulated stage -3 dB band 3.7 to 70 kHz). The output can clip at the rails at short range, which is harmless for FSK because the information is in the zero-crossings.

2.3.4 Branch A: MFB band-pass (U2B), f0 = 12.25 kHz, Q = 1.5, H0 = -2 Design with C1 = C2 = C = 2.2 nF and $\omega 0 = 2\pi \cdot 12\,247$:

- R3 = $2Q/(\omega 0 C)$ = 3 / 1.693e-4 = 17.7 k -> **18 k**
- R1 = $Q/(\omega 0 C \cdot H0)$ = 4.43 k -> **4.3 k** (E12: 4.7 k)
- R2 = $Q/(\omega 0 C(2Q^2 - H0))$ = 1.5/(1.693e-4·2.5) = 3.54 k -> **3.6 k** (E12: 3.9 k)

Ref	Value	Connection
R1BP	4.3 k	GOUT -> node MID
R2BP	3.6 k	MID -> VREF
C1BP	2.2 nF C0G/film	MID -> U2 pin 6
C2BP	2.2 nF C0G/film	MID -> U2 pin 7 (BPOUT)
R3BP	18 k	pin 7 <-> pin 6
U2B	MCP6022 half B	pin 5 = VREF
CXR	100 nF	BPOUT -> XR2211 pin 2

-3 dB band = $f0[\sqrt{(1 + 1/4Q^2)} \pm 1/(2Q)]$ = **8.8 to 17.0 kHz**. Simulated: 9.0 to 16.8 kHz, peak 12.30 kHz. 10 kHz and 15 kHz are both down only 1.4 dB and are equal, so the PLL sees no amplitude bias. With the E12 substitutes (4.7 k / 3.9 k) f0 moves to about 11.7 kHz, which is still fine.

2.3.5 Branch B: anti-alias filter to the ESP32 ADC (U3A on 3.3 V)

Ref	Value	Connection
CC2	1 μF	GOUT -> AIN
RA1 / RA2	2 k / 1 k	AIN -> AX -> VB2 (1.25 V). This /3 attenuator keeps a 5 V-pp clipped signal inside the 0.15 to 2.45 V ADC window.
RSK1, RSK2	2.0 k, 2.7 k	AX -> S1 -> S2 (unity-gain Sallen-Key; the 667 Ω source adds to RSK1)
CSK1	1.5 nF	S1 -> U3A output (feedback cap)
CSK2	680 pF	S2 -> VB2
U3A	MCP6022 (VDD = 3.3 V), follower	+IN = S2, OUT = -IN = ADCBUF
RADC / CADC	100 Ω / 1 nF	ADCBUF -> **GPIO36 (VP, ADC1_CH0)** with 1 nF to GND. Supplies the SAR sampling-cap charge.

f_c $\approx 1/(2\pi\sqrt{(R1R2C1C2)}) \approx 50$ kHz (simulated -3 dB at 49.5 kHz). At 30 kHz it is -0.4 dB; at 100 kHz -15 dB, at 200 kHz -35 dB (fs = 200 kS/s, Nyquist 100 kHz). Use **ADC1**, not ADC2: ADC2 is unusable while Wi-Fi is on. Sampling uses the ESP-IDF `adc_continuous` (DMA) driver. The classic ESP32 accepts up to about 2 MS/s nominal; 200 kS/s is used for margin and better ENOB.

Goertzel parameters (for the firmware folder):

Mode	Tones	Rb	fs	N = fs/Rb	bins k = f·N/fs	coeff = 2cos(2πk/N)
B1 (comparison)	10 / 15 kHz	5 kbps	200 kS/s	40	2, 3	1.8478, 1.6180
B2 (safe)	10 / 15 kHz	2.5 kbps	200 kS/s	80	4, 6	1.9754, 1.9021
B3 (10 kbps)	20 / 30 kHz	10 kbps	200 kS/s	20	2, 3	1.6180, 1.1756

2.3.6 Op-amp trade-off (what fits this board)

Op-amp	GBW	Input bias	Noise	Rails / supply	Package	India availability (seen)	Verdict
MCP6022	10 MHz	1 pA	8.7 nV/$\sqrt{}$Hz	RRIO, 2.5 to 5.5 V	DIP-8 dual	Robu DIP Rs 211 (older listing showed in stock); SOIC Rs 171 (out of stock when checked)	**Recommended**: one part for every stage, works on 3.3 V and 5 V
AD8605	10 MHz	0.2 pA	8 nV/$\sqrt{}$Hz	RRIO, 2.7 to 5.5 V	SOT-23-5 single	Robu Rs 88 (out of stock when checked)	Equally good electrically; needs SOT adapters (×5)
OPA380	90 MHz	3 pA	5.8 nV/$\sqrt{}$Hz	RRIO, 2.7 to 5.5 V	MSOP/SOIC	not found at the four target stores	Overkill below 100 kHz; consider only for tier C
LM358	~0.7 to 1 MHz	20 to 45 nA	~40 nV/$\sqrt{}$Hz	output only reaches V+ - 1.5 V; 0.3 V/μs	DIP-8	Rs 10 to 20 everywhere	OK only for the 1.2/2.2 kHz bring-up; the ×101 stage would have about 10 kHz bandwidth, so **not for 10 to 30 kHz**

Alternative detectors: * **OPT101**: internal 1 MΩ, 14 kHz bandwidth. It saturates at about 4.5 μA, i.e. an ordinary

lit room, and costs Rs 1,150 to 1,550 in India. Not recommended. * **BPW21**: has a built-in eye-response filter that rejects sunlight IR, but its ~580 pF capacitance limits RF·C bandwidth. Worth trying only if sunlight IR dominates.

2.4 XR2211 FSK demodulator (branch A output) Datasheet design equations (MaxLinear/Exar XR-2211A rev 1.04): * f0 = $\sqrt{}$(f1·f2) * C0 = 1/(R0·f0) * R1 = 2·R0·f0 / (f1 - f2) * C1 = 1250·C0 / (R1·ζ^2), with ζ = 0.5 * RF $\geq$ 5·R1, RB $\geq$ 5·RF * Rsum = (RF + R1)·RB / (R1 + RF + RB) * CF = 0.25 / (Rsum·Baud) * V_REF = V_CC/2 - 0.65 V (1.85 V at 5 V supply)

Main plan: 10 / 15 kHz * f0 = $\sqrt{}$(150e6) = **12 247 Hz** * R0 = 15 k + 10 k multiturn trimpot (nominal 20.9 k) * C0 = **3.9 nF** film/C0G * R1 = 2·20.9 k·12 247 / 5000 = 102 k -> **100 k** * C1 = 1250·3.9 n / (100 k·0.25) = 195 pF -> **220 pF** (ζ = 0.47) * RF = **470 k** (4.7·R1) * RB = **2.2 M** (4.7·RF; 2 × 1 M + 220 k if the kit stops at 1 M) * Rsum = 570 k·2.2 M / 2.77 M = **453 k** * CF = 0.25/(453 k·5000) = **110 pF -> 100 pF** at 5 kbps; **220 pF** at 2.5 kbps * CD (lock-detect filter) $\geq$ 16/Δf μF with Δf = 5 kHz -> 3.2 nF minimum; use **47 to 100 nF** with RD = 470 k so the carrier-detect output doesn't chatter.

Bring-up plan: Bell-202, 1200 / 2200 Hz, 1200 bps (datasheet example): R0 = 10 k + 10 k pot, C0 = 39 nF, R1 = 51 k, C1 = 3.9 nF, RF = 270 k, RB = 1.2 M, CF = 1 nF. Drive pin 2 directly from the function generator (FSK mode, 100 mV rms). The 12.25 kHz band-pass would block these tones.

XR2211 pin	Connect (main plan)
1 VCC	+5 V, 100 nF + 10 μF to GND (4.5 to 20 V allowed)
2 INP	from CXR 100 nF (internal 20 k). 10 mV to 3 V rms; limits above about 2 mV rms
3 LDF	RD 470 k
4 GND	GND
5 LDOQN	5.1 k pull-up to 3.3 V (optional "no carrier" LED)
6 LDOQ	5.1 k pull-up to 3.3 V -> ESP32 **GPIO4** (carrier detect). Datasheet Fig. 11 alternative: tie pin 6 to pin 7 so data is forced low without a carrier.
7 DO	5.1 k pull-up **to 3.3 V** (open collector, so it is 3.3 V-safe) -> ESP32 **GPIO16 (UART2 RX)**; RB 2.2 M from pin 7 to pin 8 (hysteresis)
8 COMP I	RF 470 k from pin 11; CF 100 pF to GND (datasheet Fig. 10 shows CF on pin 8; check your copy for GND vs V_REF return)
9 NC	,
10 VREF	100 nF to GND (mandatory); C1 220 pF from pin 11 to pin 10
11 LDO (loop phase detector out)	R1 100 k to pin 12; RF to pin 8; **scope this pin**: it is the analog demodulated data
12 TIM R	R0 = 15 k + 10 k trimpot to GND
13, 14 TIM C	C0 3.9 nF between them

Tuning: send alternating 0/1 (0x55 at 5 kbps) and trim R0 until pin 7 shows a 50 % duty cycle (datasheet procedure). Output polarity: pin 7 is high for the low tone (10 kHz = mark = 1), which matches UART idle.

Availability warning. No XR2211 listing was found at robu.in, robocraze.com, electronicscomp.com or evelta.com; the electronicscomp search returned nothing. Tayda lists it at $1.75 but was out of stock, and other listings are overseas brokers. Plan on importing one (LCSC, Tayda, eBay; about Rs 150 to 250 plus shipping) or using the fallback below. The part is marked obsolete by MaxLinear.

2.5 Fallback demodulator: 2 × LM567 (Rs 24 to 29 each in India) Each LM567 detects one tone. * f0 = 1/(1.1·R1·C1), with C1 = 10 nF: * 10 kHz: R1 = 9.1 k (8.2 k + 2 k trimpot) * 15 kHz: R1 = 6.1 k (5.6 k + 1 k trimpot) * Bandwidth $\approx$ 1070·$\sqrt{}$(V_in / (f0·C2)) %, with V_in $\leq$ 200 mV rms and C2 in μF. For 10 % at 10 kHz: C2 $\approx$ 0.22 μF. Output filter C3 $\approx$ 2·C2 = 0.47 μF. * Pins: 1 C3, 2 C2, 3 input ($\leq$ 200 mV rms, so add a 10:1 divider from BPOUT), 4 +5 V, 5 R1, 6 C1, 7 GND, 8 open-collector output (10 k pull-up to 3.3 V). * The ESP32

reads both outputs; a bit = (15 k detector active) ? 0 : 1. * Detection needs about 10 to 20 cycles, so expect only **300 to 500 bps**. It works as a demo, but not for the comparison.

2.6 Tier C component values

Block	Values
TX	AO3400 on a SOT-23 adapter, RG 22 Ω, RGS 100 k. LED: 5 mm high-brightness red or blue, 40 mA: R = (5 - 2.0 - 0.05)/0.04 ≈ 75 Ω for red (blue: (5 - 3.1)/0.04 ≈ 47 Ω). Pre-emphasis: 1 nF across R (τ = 75 ns) overdrives the edges.
RX TIA	BPW34 at 2.5 V (or a separate 12 V for Cd ≈ 12 pF), RF 10 k, CF 4.7 pF. f_-3dB = $\sqrt{}$(GBW/(2π·RF·C_in)) ≈ $\sqrt{}$(10 M/(2π·10 k·35 p)) ≈ 2.1 MHz; 1/(2π·10 k·4.7 p) = 3.4 MHz, so about 2 MHz overall.
Gain + comparator	1 nF/10 k AC coupling (16 kHz HPF) -> MCP6022 ×5 (BW ≈ 2 MHz) -> **LM311** (open collector, 1 k pull-up to 3.3 V, about 200 ns response). Hysteresis: 1 k input divider + 100 k positive feedback ≈ 30 mV.
Decoding	FSK 100/150 kHz @ 50 kbps: ESP32 RMT-RX or PCNT measures the period of each cycle (6.7 μs vs 10 μs). OOK Manchester @ 250 to 500 kbps: ESP32 UART + software Manchester.
Expected	2× to 10× faster than tier B. The limits are the white LED's phosphor and the LM311 edge rate; a TLV3501 (4.5 ns) is a drop-in upgrade if you can source it.

2.7 Link budget (Lambertian LOS), computed in `sim/link_budget.py` The LOS DC channel gain gives received optical power:

P_rx = P_t·(m+1)/(2πd^2)·cos$^\wedge$m(ϕ)·A·cos(ψ) (with ψ within the PD field of view)

Assumptions: * P_t(on) = 0.25 W optical (1 W white LED at 0.27 A, about 30 % efficient) * m = 1 (60° half-power angle) * A = 7.5 mm^2 * R = 0.30 A/W * The square-wave fundamental is ($\sqrt{}$2/π)·I_on rms * Noise: PD shot + servo-transistor shot of the ambient current, RF thermal, and signal shot, in the simulated branch-A noise bandwidth **B_eq = 11.3 kHz** * Non-coherent FSK: BER = $\frac{1}{2}$·exp(-Eb/2N0)

d (m)	angle	P_rx peak (μW)	I_on (nA)	I_sig rms (nA)	XR2211 input (mV rms)	ambient (μA)	noise (pA rms)	SNR (dB)	Eb/N0 @4.8 to 5 kbps (dB)
0.5	0°	2.39	716	322	clips at ~1.7 V (fine for FSK)	10	279	61	65
1	0°	0.60	179	81	647	2 / 10 / 50	137 / 277 / 605	55 / 49 / 43	59 / 53 / 46
2	0°	0.15	45	20	162	10	277	37	41
3	0°	0.066	20	9.0	72	2 / 10 / 50	136 / 276 / 605	36 / 30 / 23	40 / 34 / 27
3	30° tx & rx	0.050	15	6.7	54	10	276	28	31

d (m)	angle	P_rx peak (μW)	I_on (nA)	I_sig rms (nA)	XR2211 input (mV rms)	ambient (μA)	noise (pA rms)	SNR (dB)	Eb/N0 @4.8 to 5 kbps (dB)
3 + 15° TX lens	0°	2.0	597	269	clips	10	278	60	63

What the table means: * Thermal and shot noise alone predict BER $< 10^{-12}$ everywhere in the 0.5 to 3 m target, because ideal non-coherent FSK needs only 14.2 dB Eb/N0 for 10^{-6}. * Estimated maximum range at 10 kbps, keeping 10 dB implementation margin: about **6 m (dim, 2 μA), 4.4 m (10 μA), 3.0 m (50 μA near a window)** with the bare LED. * **In practice the link will be limited by interference, not noise:** * harmonics of lamp flicker * CFL and electronic-ballast tubes switching at 20 to 60 kHz, right next to the 20/30 kHz tones * PWM-dimmed LED panels at 1 to 20 kHz * misalignment, PLL lock range, and bit-timing slips * This is why section 5 measures the ambient spectrum first. * The XR2211 input level stays above its 10 mV rms threshold out to about 7 m, so no AGC is needed.

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3. Netlists and simulation results

- **Simulator:** ngspice 44.2 ran in the sandbox: Debian `.deb`s extracted to user space, since there was no root access. Results below are from it.
- **LTspice:** both `.cir` files are plain SPICE (`.subckt`, `.model ... VDMOS`, `.ac` / `.tran`) and open directly. Plot `V(bpout)`, `V(adc)`, `V(tout)`; with AC 1 A in `IPH`, volts = ohms of transimpedance.
- **ngspice on your PC:** `cd sim && ngspice -b rx_ac.sp && ngspice -b tx_tran.sp && python3 make_plots.py && python3 link_budget.py`.
- **Models:** the op-amp is a single-pole macro (A0 = 100 dB, GBW = 10 MHz, 6 pF inputs). It has no slew limit, clipping or voltage noise, so swap in Microchip's MCP6022 SPICE model for large-signal checks. The IRLZ44N and AO3400 VDMOS models are approximate fits to datasheet headline numbers, and the LED is a diode fitted to V_F = 3.2 V at 0.3 A.

RX AC results (ngspice, `rx_ac_response.png`):

Node	Midband	-3 dB band	Key points
TOUT (TIA + servo)	93.4 dBΩ = 46.8 kΩ	153 Hz to 103.5 kHz	100 Hz: -5 dB (servo)
GOUT (×101)	133.2 dBΩ = 4.6 MΩ	3.7 to 70 kHz	
BPOUT (to XR2211)	**139.6 dBΩ = 9.5 MΩ peak at 12.30 kHz**	**9.0 to 16.8 kHz**	10 kHz: 138.1 dBΩ, 15 kHz: 138.1 dBΩ (equal, -1.5 dB); **100 Hz: 34.9 dBΩ -> 105 dB rejection**; 1.2 kHz: -37 dB
ADC (to ESP32)	123.7 dBΩ = 1.53 MΩ	3.7 to 49.5 kHz	20 k: 123.7, 30 k: 123.2 dBΩ; 100 Hz: -61 dB; 100 kHz: -15 dB; 200 kHz: -35 dB

- DC operating point with 10 μA ambient: TOUT 2.500 V (the servo removed the DC exactly), U1B output 0.62 V, V(RE) = 0.100 V (= 10 μA), all stages centred on their bias.
- Noise (`rx_noise.png`): circuit noise at BPOUT is 1.30 mV rms (100 Hz to 100 kHz), equivalent to 136 pA rms at the input in band, which includes Q1's shot noise. Add the photodiode's own ambient shot noise for the totals in section 2.7.

TX transient (`tx_transient.png`, 15 kHz tone):

Quantity	Value
LED on-current	267 mA
V_DS(on)	9 mV
Off leakage	2.7 μA
Gate rise (10 to 90 %)	0.92 μs (Miller plateau)
Drain fall	82 ns
Turn-on delay	0.21 μs
Off-state drain	~2.8 V (LED sub-threshold)

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4. Bill of materials (INR, incl. GST where shown)

Prices are what the linked page or search listing showed in Oct 2026; stock changes daily. Every URL below was seen via search or fetch. **"No URL"** means none was verified, so buy it from a local shop or any kit.

4.1 Core: tiers A + B on one RX board

#	Part	Qty	Unit Rs	Line Rs	Store URL (seen)	Note
1	ESP32 38-pin DevKit (WROOM-32)	2	422	844	link	TX + RX. Cheaper option: Rs 340 at link
2	MCP6022-I/P dual op-amp DIP-8	3	211	633	link	U1, U2, U3. SOIC alternative Rs 171 (OOS): link
3	XR2211 FSK demodulator	1	~200 (import)	200	**not stocked at the four Indian stores**; Tayda (OOS): link	Order early, or use row 4
4	LM567 tone decoder (fallback)	2	29	58	link	Optional if the XR2211 arrives
5	BPW34 PIN photodiode	2	59	118	link	Also Rs 64: link ; Rs 92: link
6	IRLZ44N logic-level MOSFET	2	44	88	link	
7	BC547 NPN (servo Q1)	2	~9	9	link	Listing price Rs 8.5 (pack option)
8	1 W white LED with star heatsink	2	13	26	link	
9	6.8 Ω 2 W metal-oxide resistor	2	14	28	link	Showed out of stock; any 5.6 to 8.2 Ω 2 W from a local shop

#	Part	Qty	Unit Rs	Line Rs	Store URL (seen)	Note
10	Metal-film resistor kit, 600 pcs, 30 values, 1 %	1	174	174	link	E24 values (4.3 k, 3.6 k) may need series combos
11	Ceramic capacitor kit, 30 values, 2 pF to 0.1 μF	1	188	188	link	Use C0G/film for C0, CF, BPF caps where possible
12	3296W 10 k multiturn trimpot	3	18	54	link	XR2211 R0, LM567, spare
13	MB102 830-point breadboard	2	64	128	link	TX and RX separate
14	Perfboard 10 $\times$ 15 cm	2	49	98	link	Final RX build (robu Rs 99, OOS: link)
15	20 mm 15° TIR LED lens	2	15	30	link	One on TX, one reversed over the PD as a $\pm 7.5°$ concentrator
16	5 V 2 A adapter (TX LED rail)	1	183	183	link	Listed as “5V 2A Power Adapter Rs 183”. Skip it if a 2 A phone charger plus USB cable powers the TX ESP32 VIN.
17	Electrolytics (10 μF $\times$6, 470 μF $\times$1), jumper wires, 2 $\times$ 8-pin + 1 $\times$ 14-pin DIP sockets	1 lot	~80	80	no URL	Local shop

Core total (tiers A + B)	**≈ Rs 2,939**		Rs 2,698 without the optional adapter and LM567s. Add Rs 0 to 150 shipping; free above Rs 999 per store at Robu and Robocraze, so consolidate orders.

4.2 Optional and tier C add-ons

Part	Qty	Rs	URL (seen)	Use
AO3400 SOT-23 MOSFET	5	8 ea	link	Fast switch for tiers B and C
SOT-23 to DIP adapter (7Semi)	2	29	link (also link)	For the AO3400
LM311 comparator DIP-8	1	19	link (robu Rs 17: link)	Tier C slicer
XR2206 signal-generator kit (soldered)	1	330	link	Analog TX option; needs 9 to 12 V. The bare IC is Rs 595 to 729 in India, so the kit is cheaper.
12 V 2 A adapter	1	185	link	Only for the XR2206
Convex glass lens set (50 mm, f 15 to 30 cm)	1	79	link	High-gain RX optics for long-range tests (FOV only about ±0.5°)
OPT101 module	,	1,150 to 1,769	link	Not recommended (section 2.3.6)
AD8605 SOT-23 op-amp	,	88	link	Alternative to the MCP6022

---

5. Test and measurement plan

5.1 Stage-by-stage bring-up (scope at each point)

Step	Stimulus	Probe	Expect	Pass criterion
1. XR2211 alone	Function generator in FSK mode: 1200/2200 Hz, 1200 bps, 100 mV rms into pin 2 (Bell-202 values)	pin 11 (analog), pin 7 (data) vs FG trigger	Clean square at pin 7 that copies the FG's keying	Duty within 50 ± 5 % on 0x55 after trimming R0
2. Main XR2211 values	FG FSK 10/15 kHz, 5 kbps, 100 mV rms	pin 11, 8, 7	Pin 11 eye open, pin 7 = data	Correct bits at 10 mV to 1 V rms input

Step	Stimulus	Probe	Expect	Pass criterion
3. TX driver	ESP32 tone 10 kHz	CH1 gate, CH2 drain; LED current = V across R_LED / 6.8	Gate rise ≈ 1 μs, drain fall < 0.2 μs, I_on 0.25 to 0.3 A	Matches `tx_transient.png`; MOSFET cold to touch
4. TIA DC	no TX, room light	TOUT, V(RE)	TOUT = 2.50 V ± 20 mV (servo on); V(RE)/10 k = ambient current	Log V(RE) for every later experiment
5. TIA AC	TX at 1 m, 10 kHz	TOUT (AC-coupled, 1 mV/div)	~8 mV p-p square (179 nA × 47 k), plus 100 Hz ripple	No oscillation: scope at 1 MHz span shows nothing near 1 to 5 MHz
6. Gain stage	same	GOUT	~0.8 V p-p, rounded square	Not clipping at 1 m; clipping at 0.5 m is OK
7. Band-pass	same	BPOUT, scope FFT	Sine at 10 or 15 kHz; 100 Hz component below the noise	FFT: 100 Hz at least 60 dB under the tone
8. Front-end Bode	FG sine sweep 100 Hz to 200 kHz into the gate (square drive; read the fundamental) or, better, FG -> 100 kΩ -> TIN for electrical injection (10 μA/V)	BPOUT, ADC node	Compare with `rx_ac_response.png`	Peak within ±10 % of 12.3 kHz; within ±2 dB of simulation
9. ADC branch	TX at 1 m	ADCBUF ($\leq$ 3.3 V!), ESP32 raw dump	1.25 V centred, $\leq$ 2.4 V p-p	Goertzel bins separate by > 20 dB

5.2 BER test method

1. The TX ESP32 sends framed **PRBS-15** ($x^{15} + x^{14} + 1$) as 8N1 UART bytes. Frame: preamble 0x55 ×8, sync word 0xD3 0x91, 64-byte payload, CRC-16.
2. The RX ESP32 runs the same PRBS generator. After sync it XOR-compares received bits, counting bit errors, frame errors and lost syncs. It prints `d, angle, lux/V(RE), Rb, demod, bits, errors` over USB serial to a laptop CSV logger.
3. Do both demodulators in the same run: the XR2211 path on UART2 and the Goertzel path on the ADC see identical light.
4. Bits per point: run until 100 errors or 10^7 bits (about 33 min at 5 kbps, 17 min at 10 kbps). With zero errors in N bits, report **BER < 3/N** (95 % confidence).
5. Eye diagrams: XR2211 pin 11 and the Goertzel decision metric ($|X1|^2$ - $|X0|^2$) with scope persistence, triggered on the TX bit clock (spare GPIO).

5.3 Experiments and expected plots

Experiment	Sweep	Fixed	Expected plot / result
Range	d = 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 5 m (tape on the floor)	0°, 5 kbps, room LED tubes	BER vs d (log y): floor below 10^{-6} to about 3 to 4 m, then a cliff. Overlay the received tone amplitude at BPOUT (should fall as $1/d^2$) and the link-budget curve.
Angle	TX tilt and RX tilt, 0 to 60° in 10° steps	1 m and 2 m	Amplitude follows cosˆm(ϕ)·cos(ψ) with m ≈ 1; BER cliff near 50 to 60°. With lenses: a sharp cliff at ±7.5°.
Ambient	dark / room LED / CFL / tube / window daylight / phone flashlight as interferer	2 m	V(RE)-measured ambient on x, BER and tone SNR on y. Expect graceful loss with DC (servo) but **sharp loss with CFL/ballast interference near 20 to 60 kHz**, which hits tier B3 more than 10/15 kHz.
Ambient spectrum	none (TX off)		Scope FFT of GOUT, 0 to 100 kHz, for each light source. This is the data for adaptive tone selection.
Bit rate	1.2 (Bell), 2.5, 5, 7.5, 10 kbps	2 m	BER vs Rb for PLL vs Goertzel. Expect the XR2211 to degrade above about 5 kbps on 10/15 kHz (loop settling), and the Goertzel path to hold to 5 kbps on 10/15 kHz and to 10 kbps on 20/30 kHz.
PLL vs Goertzel	same light, same frames	all above	Paired BER table and a cost/complexity comparison: the paper figure.
Throughput demo	text and small file transfer	1 m	Goodput vs raw rate (framing/CRC overhead)

5.4 Measurements to record per point Distance, angles, V(RE) (ambient), tone amplitude at BPOUT (mV rms), noise floor at BPOUT with TX off, BER per demod, XR2211 lock flag (pin 6) duty, LED current, room light type, time.

---

6. Common pitfalls and debugging checklist

Optics and light - [ ] PD reversed: with the cathode not at +5 V, TOUT rises with light and the servo runs away. Check the multimeter diode test. - [ ] Direct sun or a desk lamp shining on the PD saturates even the servo (V(RE) > 4 V). Add a 2 to 3 cm black-tube shade (a pen barrel works) or move it. - [ ] CFL, tube-light ballasts or PWM-dimmed panels inject 20 to 60 kHz lines: look at the GOUT FFT with TX off. If a line lands near a tone, move the tones (keep $\Delta f = Rb$) or switch off that lamp. Note it in the paper. - [ ] Phone cameras show the LED

flickering at 10 kHz (rolling shutter). That's normal and not visible to the eye. - [ ] A lens on the TX makes the link alignment-critical: align it with the TX running a 10 kHz tone while watching BPOUT amplitude.

Analog - [ ] TIA oscillates (MHz ringing at TOUT, scope at 1 μs/div): CF is missing or too small (< 4 pF), or there is a long wire at TIN. Keep the PD leads under 1 cm and put RF/CF right on the op-amp pins. - [ ] Huge 100 Hz at GOUT: the servo isn't working (check Q1 pinout C-B-E and integrator polarity: U1B (+) = VREF, (-) via RI from TOUT). Another cause is VREF ripple: decouple VREF with 10 μF at each op-amp. - [ ] Gain stage drifts or latches: CG must be in series with RG1 to VREF (not to GND), or the $\times$101 gain amplifies the offset and DC. - [ ] Band-pass peak at the wrong frequency: X7R ceramics lose capacitance with temperature and voltage. Use C0G or film for 2.2 nF, 3.9 nF and 220 pF. - [ ] MCP6022 on 5 V feeding the ESP32 ADC directly will kill the pin. Only U3 (on 3.3 V) may touch GPIO36. - [ ] Ground loops: the TX and RX ESP32s are both on laptop USB, so the shared ground carries LED current and noise. Power the TX from a separate charger.

XR2211 / PLL - [ ] No output: pin 10 needs 100 nF (mandatory per the datasheet); input must be AC-coupled; V_CC $\geq$ 4.5 V. - [ ] Output stuck high or low: f0 is mistuned. Send 0x55 and trim R0 for 50 % duty, and check pin 11 swings around V_REF. - [ ] Bit errors at 5 kbps but fine at 2.5 kbps: CF too large (data filter too slow) or C1 too large (loop too slow). Try CF 68 to 100 pF; check $\zeta \approx 0.5$. - [ ] Polarity inverted: pin 7 is high for the low tone. Either swap the tone map at TX or invert in the ESP32 UART (`uart_set_line_inverse`). - [ ] Pull-up to 5 V on pin 7 feeding the ESP32 exceeds 3.6 V abs max: pull up to 3.3 V only.

TX / firmware - [ ] LED flashes at ESP32 boot: RGS 100 k is missing, or a strapping pin is used for the gate. - [ ] Frequency glitches at bit boundaries (spectral splatter, PLL slips): use RMT/timer phase-continuous switching, not repeated `ledcChangeFrequency`. - [ ] The UART start bit is lost because the TX emits no carrier between frames: keep the carrier on (idle mark at 10 kHz) and send a 0x55 preamble. - [ ] Goertzel bins mismatched: fs drifts from the nominal 200 kS/s. Measure the real rate by timing 10^6 samples; use non-integer-k Goertzel if needed. - [ ] Wi-Fi on and ADC2 used: ADC2 is blocked by Wi-Fi. Always use ADC1 (GPIO32 to 39).

Measurement hygiene - [ ] Scope ground lead pickup at 10 kHz: use the short ground spring on BPOUT/TOUT. - [ ] Record V(RE) (ambient) with every BER point; without it the ambient experiments can't be compared. - [ ] Repeat each point three times, at different times of day, because daylight varies.

Notes, assumptions, open items

- The link budget assumes 0.25 W optical from the 1 W LED and 0.30 A/W effective BPW34 responsivity for white light. Both are estimates ($\pm$3 dB). Measure TOUT at 1 m to calibrate: expect about 8 mV p-p.
- The simulation op-amp is an ideal-ish macro. Large-signal clipping, slew rate and op-amp voltage noise aren't modelled; at these frequencies they are negligible, apart from intentional clipping at short range.
- The XR2206 pin assignments and the XR2211 CF/C1 return pins come from the datasheet figures as described. Check them against your printed datasheet before soldering.
- Prices and stock are as seen during the search in Oct 2026. Several low-cost listings (MCP6022 SOIC, AO3400 alternatives, 6.8 Ω 2 W) were out of stock at that moment.

Part D: Firmware, Simulation and Code

Li-Fi BFSK link: firmware, simulation and PC tools

A visible-light link for a student project. An LED sends binary FSK tones (bit 0 = f0, bit 1 = f1, 50 % duty square waves). A photodiode front-end feeds an ESP32 ADC, and the ESP32 recovers the bits with Goertzel filters. An Arduino Uno version (timer-generated tones, XR2211 PLL demodulator) is included too.

Defaults: **f0 = 10 kHz, f1 = 15 kHz, 1000 bit/s, receiver fs = 48 kSps**.

```
PC --USB-- TX (ESP32 or Uno) -- MOSFET -- LED  ~~~ light ~~~  photodiode -- TIA -- HPF/gain -- RX ADC (ESP32)
  tools/                     tx_esp32/ tx_arduino_uno/                                        rx_esp32/
                                                                          +-- XR2211 -- data line -- Uno rx_arduino_uno_xr2211/
```

Repository layout

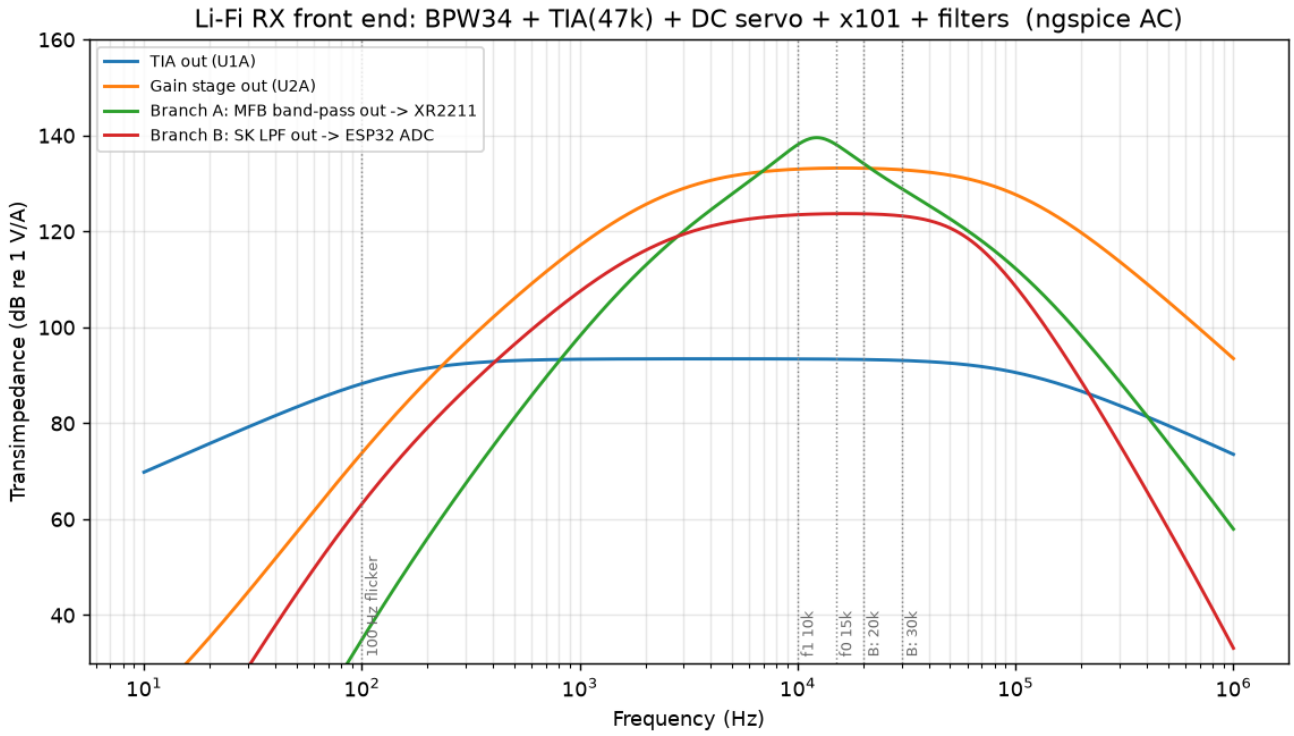


Figure 1: RX front-end AC response (ngspice)

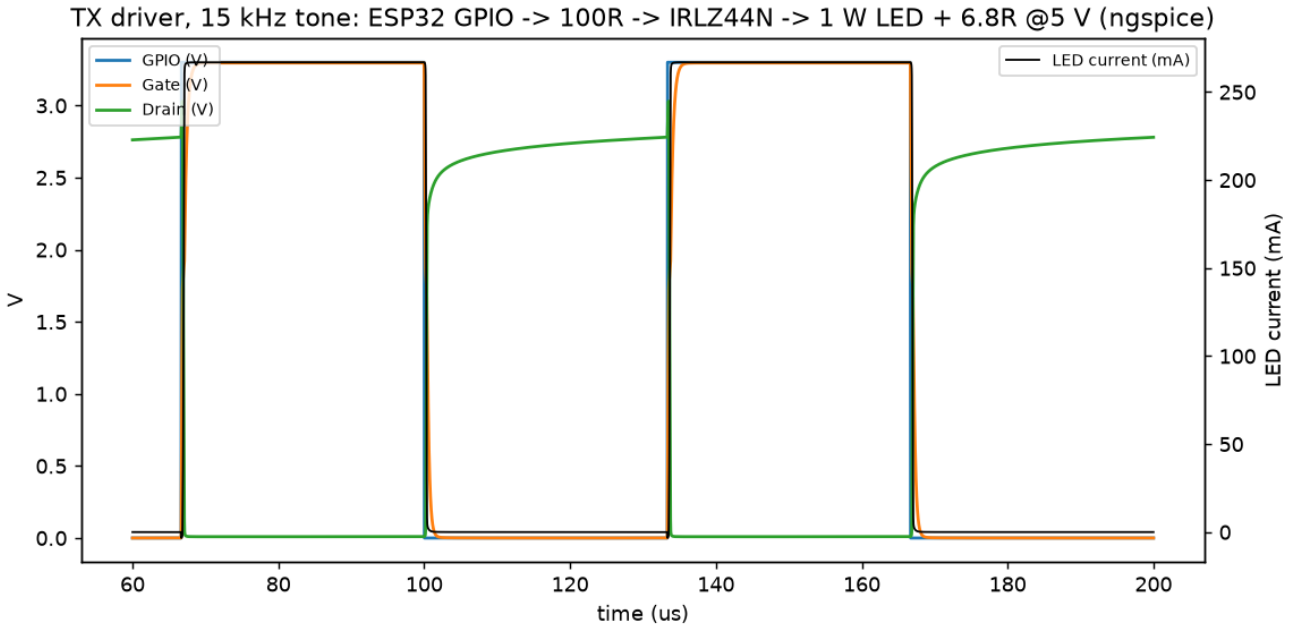


Figure 2: TX driver transient (ngspice)

Path	What it is
`common/lifi_proto.h`	Framing, CRC-16-CCITT, Hamming(7,4), PRBS9, streaming TX framer and RX deframer (plain C, header-only)
`common/lifi_demod.h`	Streaming demodulator: DC/flicker removal, 8-phase Goertzel bank, preamble timing lock, tracking, RSSI/SNR
`tx_esp32/`	ESP32 transmitter (Arduino-ESP32 core 3.x). Uses the DDS timer ISR for continuous-phase FSK, or LEDC
`rx_esp32/`	ESP32 receiver. Continuous (DMA) ADC, Goertzel demod, CRC, BER statistics
`tx_arduino_uno/`	Uno transmitter. Timer1 continuous-phase toggling, or `tone()`. Packet or plain-UART coding
`rx_arduino_uno_xr2211/`	Uno receiver behind an XR2211. 8x oversampled DPLL on the data line, or SoftwareSerial in UART mode
`sim/lifi_ref.py`	Python reference: same framing and demod as the C code, plus a filter-bank receiver and the XR2211-line DPLL
`sim/channel.py`	LED/front-end low-pass, Lambertian distance loss, ambient DC, 100 Hz flicker, AWGN, 12-bit ADC, clock offset
`sim/run_sim.py`	Full simulation. Writes `sim/out/*.png` and `results.json`
`tests/`	pytest suite (47 tests). `tests/host/host_harness.cpp` builds the **firmware headers on the PC**
`tools/lifi_tool.py`, `tools/lifi_gui.py`, `tools/lifi_link.py`	PC CLI and Tk GUI (pyserial): send text and files, run BER tests, live BER/RSSI plot
`tools/sync_common.py`	Copies `common/*.h` into each sketch folder. Arduino can't include files from outside the sketch, so each sketch carries a copy. A test checks the copies match

Frame format

```
| 0xAA 0xAA 0xAA 0xAA | SYNC (16 b) | LEN (8 b) | PAYLOAD (0..255 B) | CRC16 (16 b) |      bits MSB first
```

- `SYNC = 0x2DD4`: plain body. `SYNC = 0xD263`: LEN, PAYLOAD and CRC are Hamming(7,4) coded (each byte becomes two 7-bit codewords, ordered p1 p2 d1 p3 d2 d3 d4). The receiver picks the mode from the sync word, so coded and plain frames can be mixed on the same link.
- CRC-16/CCITT-FALSE (poly 0x1021, init 0xFFFF) over LEN and PAYLOAD. Check value: "123456789" gives 0x29B1.
- The two sync words differ in 14 bits. Every 16-bit window of preamble+sync is at least 5 bits away from either sync word.
- Between frames the TX idles on the f1 tone ("mark"). This keeps the LED's average brightness constant, so there's no visible flicker.
- BER-test frame: `0xB5, seq, 32 bytes PRBS9`. The RX counts bit errors in the PRBS part.

Receiver algorithm (`common/lifi_demod.h`)

1. **Ambient light**: two cascaded running-mean trackers (alpha = 1/16, about 500 Hz corner at 48 kSps) remove the DC level and most of the 100 Hz mains flicker. The Goertzel window rejects what's left (in simulation the 100 Hz line drops by about 29 dB).
2. **8 timing phases**: each phase has its own Goertzel pair (f0, f1) over one bit window, offset by 1/8 bit. Fractional samples-per-bit are handled, so the measured ADC rate (for example 47 993 Sps) can be used directly.
3. **Decision**: bit = P(f1) > P(f0). Margin m = (P1-P0)/(P1+P0).
4. **Acquisition**: every phase looks for `[0xAA][SYNC]`. On a match, the phase with the best average |m| over the 0xAA preamble is locked. A phase that is later than the matching one skips one window, so bit alignment is

kept.

5. **Tracking**: after each bit, the lock can step to a neighbouring phase whose slow-average |m| is clearly better. This absorbs crystal offset between TX and RX (tested to ±300 ppm, and fractional fs).
6. **Per-frame statistics**: RSSI = 20·log10(mean winning-tone amplitude in ADC counts). SNR = winner/loser tone energy. Quality = mean |m|.

CPU estimate on the ESP32 (not measured): 16 Goertzel updates per sample × 48 kSps ≈ 0.8 M multiply-adds/s, a few % of one 240 MHz core.

Design note: why fs = 48 kSps and not 40 kSps At fs = 40 kSps = 4·f0, the 3rd harmonic of the square-wave f0 tone (30 kHz) folds back exactly onto 10 kHz. The f0 tone energy then swings by about 3 dB depending on the ADC sampling phase. The simulation measured Goertzel power from 24.7 to 47.3 across sampling phases, which cost about 1 dB in BER. At 48 kSps the harmonics fold to 2, 3, 18 and 21 kHz. Those are other Goertzel bins, which are orthogonal over a 48-sample window (`sim/out/spectrum.png`). If you change f0, f1 or fs, check that k·f0 and k·f1 (k = 3, 5, 7) don't fold onto either tone. Also keep the tone spacing a multiple of the bit rate (here 5·Rb), so the two tones are orthogonal.

Simulation results (`python sim/run_sim.py`, about 2 min; `--quick` takes about 20 s)

Eb/N0 is defined from the tone energy the detector actually sees. The **full chain** row runs the firmware's own `lifi_demod.h`, compiled for the PC, on 400 BER-test frames per point. The channel includes ambient DC (0.6 V), 100 Hz flicker (0.3 V peak), +100 ppm clock offset, LED (100 kHz) and front-end (30 kHz) low-pass, and a 12-bit ADC.

Eb/N0 (dB)	8	10	11	12	13	14
theory 0.5·exp(-Eb/2N0)	2.1e-2	3.4e-3	9.2e-4	1.8e-4	2.3e-5	1.8e-6
Goertzel, square tones, ideal timing (4e5 bits)	2.2e-2	3.6e-3	9.9e-4	2.2e-4	2.3e-5	-
filter bank (2×BPF + energy), sine	3.1e-2	6.0e-3	2.0e-3	5.2e-4	1.3e-4	-
full chain, plain (BER / frames lost / FER)	2.7e-2 / 40 % / 100 %	4.5e-3 / 9 % / 74 %	1.3e-3 / 2.5 % / 34 %	3.4e-4 / 1.5 % / 9 %	6.8e-5 / 0 / 1.8 %	**0 / 0 / 0**
full chain + Hamming(7,4) (Eb per info bit)	6.6e-2	8.5e-3	3.3e-3	4.4e-4	5.9e-5	9.8e-6 / FER 0.25 %

- The Goertzel receiver matches non-coherent BFSK theory. The full firmware chain, with real impairments, comes within about 0.3-0.5 dB of it. The filter bank is about 1 dB worse. Full-chain BER counts only frames whose sync was found; frame loss is listed separately.
- Hard-decision Hamming(7,4) costs 2.4 dB of energy per coded bit. It only breaks even at about 13 dB (BER ≤ 1e-4), and it also weakens the uncoded preamble and sync. Use it on a link with occasional single-bit hits, not as a range extender.
- Range: 0.5 W optical LED (60° half-angle), BPW34, 100 kΩ TIA ×10, 8 mV rms noise. Error-free (1e5 bits) to 5 m, BER 1.1e-4 at 6 m, 4.6e-3 at 7 m (`ber_distance.png`).
- Message tests (Python and firmware-C demod both decode the 55-byte message). Cases: square continuous-phase at 14 dB +150 ppm; Hamming at 12 dB -200 ppm (2 bits corrected); phase-reset tones like LEDC; sine with 0.6 V flicker.
- XR2211 path: a behavioural XR2211 (band-pass energy comparator, inverted output) feeds the Uno DPLL at 2000 ppm clock offset. Both frames decode, and the polarity is detected automatically.

Plots: sim/out/ber_curve.png, spectrum.png, waveforms.png, ber_distance.png.

Building

Install arduino-cli. Then install the esp32:esp32 core (3.x) from https://espressif.github.io/arduino-esp32/package_esp3
and arduino:avr.

```
python tools/sync_common.py           # after editing common/
arduino-cli compile --fqbn esp32:esp32:esp32 tx_esp32
arduino-cli compile --fqbn esp32:esp32:esp32 rx_esp32
arduino-cli compile --fqbn arduino:avr:uno tx_arduino_uno
arduino-cli compile --fqbn arduino:avr:uno rx_arduino_uno_xr2211
arduino-cli upload -p COM5 --fqbn esp32:esp32:esp32 tx_esp32
```

Compile-checked here (not flashed) with arduino-cli 1.5.1, esp32 core 3.3.12 and arduino:avr 1.8.8, using `--warnings all` (no warnings from the sketch code). See BUILD_LOG.md for sizes.

Hardware notes

TX driver: GPIO25 (ESP32) or D9 (Uno) -> 100 Ω -> gate of a logic-level N-MOSFET (AO3400, IRLML6344, or IRLZ44N on 5 V only), with 10 kΩ gate-to-source pull-down. Use low-side switching: LED string and current-limit resistor (or a constant-current driver) from the supply to the drain, source to ground. At 3.3 V gate drive, pick a MOSFET with R_DS(on) specified at V_GS = 2.5 V. Keep the LED driver free of large output capacitors: the light has to follow 15 kHz edges.

RX front-end (suggested): photodiode (BPW34) reverse-biased into a transimpedance amp, e.g. MCP6022 or OPA350. Use Rf = 100 kΩ and Cf ≈ 47 pF (≈ 34 kHz bandwidth, which is also the anti-alias filter). Then AC coupling (≈ 100 Hz HPF) and a ×10 stage biased at 1.65 V, into GPIO34 (ADC1). With 12 dB attenuation, the ESP32 ADC is most linear between about 0.15 and 2.45 V, so keep the swing inside that. Bright ambient light can saturate a DC-coupled TIA. Then lower Rf or shade the photodiode; the digital mean removal can't undo clipping.

XR2211 for 10/15 kHz, 1000 baud. These come from the XR-2211A datasheet design instructions (https://www.maxlinear.com/ds/xr2211av104.pdf): f_o = $\sqrt{(f1 \cdot f2)}$ = 12.25 kHz. R0 = 20 kΩ (15 k plus a 10 k trimmer). C0 = 1/(R0·f_o) ≈ 3.9 nF. R1 = 2·R0·f_o/(f2-f1) ≈ 100 kΩ. C1 = 1250·C0/(R1·ζ^2) with ζ = 0.5 ≈ 200-220 pF. RF ≥ 5·R1 -> 510 kΩ. RB ≥ 5·RF -> 2.7 MΩ. CF = 0.25/(R_sum·baud) ≈ 470 pF. Input is AC-coupled (0.1 μF), 10 mV-3 V rms. DO (pin 7) is open-collector: use 5.1 kΩ to 5 V, then Uno D2. DO is high for the lower tone, so it is inverted relative to bit = 1 -> f1. The sketch detects polarity from the sync word. Tune the trimmer for 50 % duty on DO while the TX sends the 0xAA preamble (`!bert`).

Using it

TX (115200 baud): type a line to send it. Commands: `!f0 <Hz>`, `!f1 <Hz>`, `!rate <bps>`, `!mode dds|ledc`, `!fec on|off`, `!idle mark|off|on`, `!bert <n> [gap_ms]`, `!hex <bytes>`, `!info`. * `dds` (default): a hardware timer at 160 kHz steps a 32-bit phase accumulator. Tone switches are phase-continuous, and bit timing comes from a second accumulator. Edge jitter is at most one tick (6.25 μs). Lower `TICK_HZ` if ISR load matters; keep TICK_HZ ≥ 8·f1. * `ledc`: `ledcChangeFrequency()` at each bit boundary, paced with `esp_timer`. Simpler, but the switch happens at the next PWM period and has some timing jitter. Neither mode needs Manchester coding: the sync and preamble give timing, and the receiver tracks it.

RX (921600 baud) prints `RX ok "text"` and machine-readable lines: `#FRAME,ok,fec,len,rssi_db,snr_db,quality,correct`
`#BERT,frames,bits,errors,ber,lost`, and every 2 s `#STAT,ms,fs_meas,syncs,frames,crc_fail,bert_frames,bert_bits,`
At boot it measures the real ADC rate for 1 s and uses it (`AUTO_CAL_FS`). Commands: `!reset`, `!info`.

PC tools (`pip install pyserial matplotlib`):

```
python tools/lifi_tool.py --tx COM5 send "hello over light"
python tools/lifi_tool.py --tx COM5 sendfile notes.txt --repeat 2
python tools/lifi_tool.py --tx COM5 bert --count 500
python tools/lifi_tool.py --rx COM6 monitor --plot --save-files received/ --log rx.csv
python tools/lifi_gui.py
```

Tests

```
pip install numpy scipy matplotlib pytest pyserial
python -m pytest -q tests        # 47 passed
```

The tests cover: * CRC check value, Hamming correcting every single-bit error, sync-word distances, and PRBS period * Framing round trips (0-255 bytes, plain and FEC) * **C TX framer output equals the Python framer, bit for bit** (host-compiled firmware header) * **Python demod and firmware C demod decode identical frames** * Demod under DC, flicker, ±300 ppm, phase-reset tones, FEC, and 44.1 kSps (fractional samples per bit) * RSSI scaling, BER within ±20 % of theory at 6 and 9 dB, the XR2211 DPLL decoder (±2000 ppm, inverted line), log parsing, file chunking and reassembly, and that the sketch header copies are identical The host harness is built with g++/clang++ if present, otherwise with `pip install ziglang`.

Options and stretch goals

- **Hamming(7,4)**: implemented end to end (`!fec on`; RX auto-detects it). See the trade-off above.
- **File transfer**: implemented in `tools/`. Files are split into 200-byte chunks with a header (id, seq, total, CRC16 of the chunk); chunk 0 carries the name and a whole-file CRC. The link is one-way with no ACK, so use `--repeat` and the receiver keeps the good copies. At 1 kbps the goodput is about 0.75 kbit/s.
- **Audio streaming** (not implemented): 8 kHz PCM needs 64 kbit/s. Codec2 at 1200-3200 bit/s fits after raising the link to about 4 kbit/s. For example, use f0/f1 = 40/60 kHz at 4 kbps with fs = 192 kSps on the ESP32 ADC (spb = 48, same demod), a faster TIA, a stream of fixed-length frames, and a small jitter buffer. Frame loss gets masked by repeating the last codec frame.
- Further ideas: soft-decision decoding (keep m instead of the hard bit) with a convolutional code, an ACK channel (IR LED back) for ARQ, carrier-sense on the idle tone, and Wi-Fi/BLE bridging.

Limitations (honest list)

- Nothing has been flashed or tried on hardware. The sketches compile, and the protocol and demod logic run on the PC from the same headers, but ISR timing, ADC behaviour and analog values still need checking on the bench.
- The ESP32 continuous-ADC rate is not exact. The firmware measures it at boot. The ESP32 ADC is also noisy (several LSB), which the range estimate covers only through the 8 mV noise figure.
- The DDS ISR at 160 kHz is the heaviest part of the TX. If the core misses ticks, lower `TICK_HZ` or switch to `ledc`.

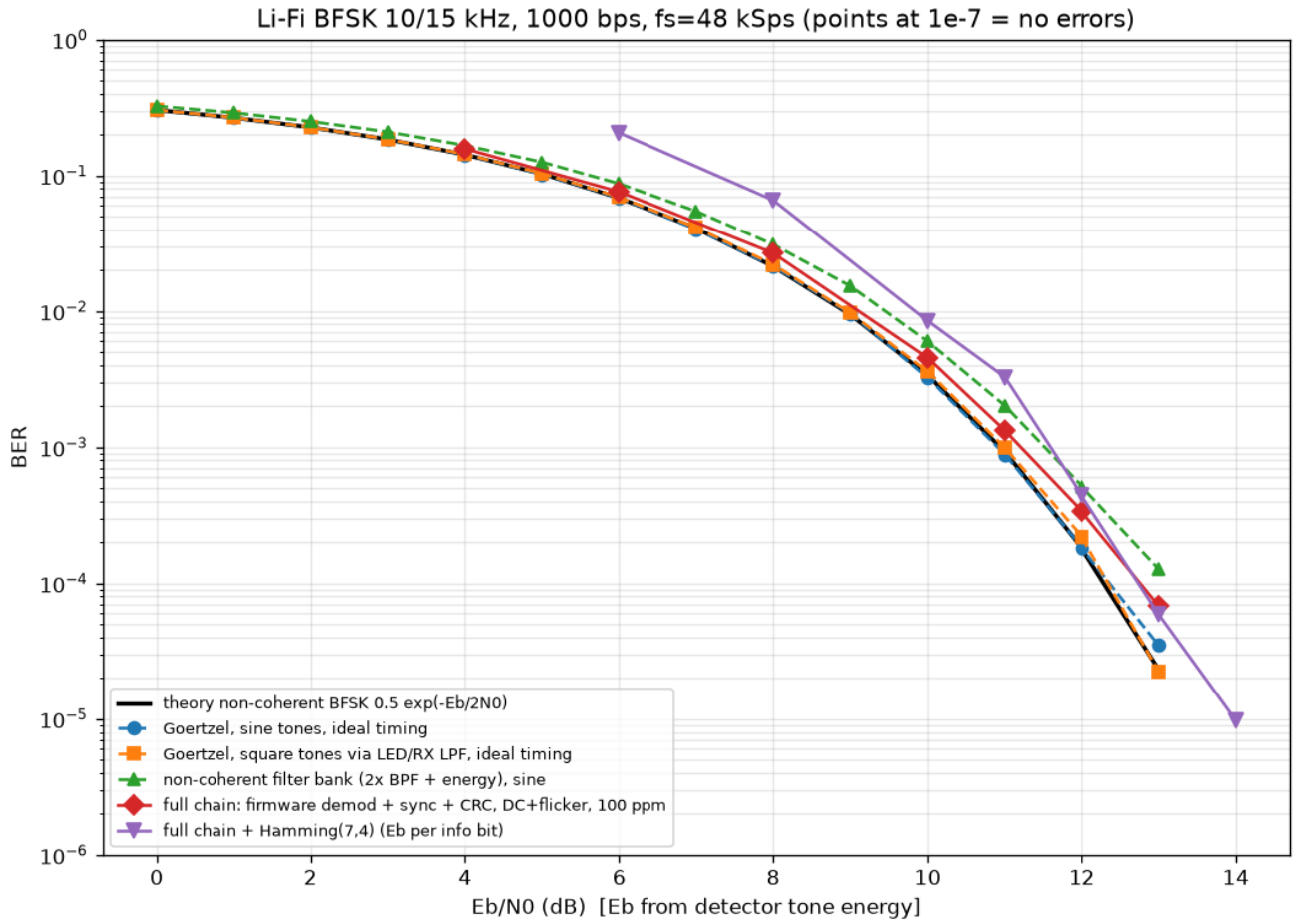


Figure 3: BER vs Eb/N0

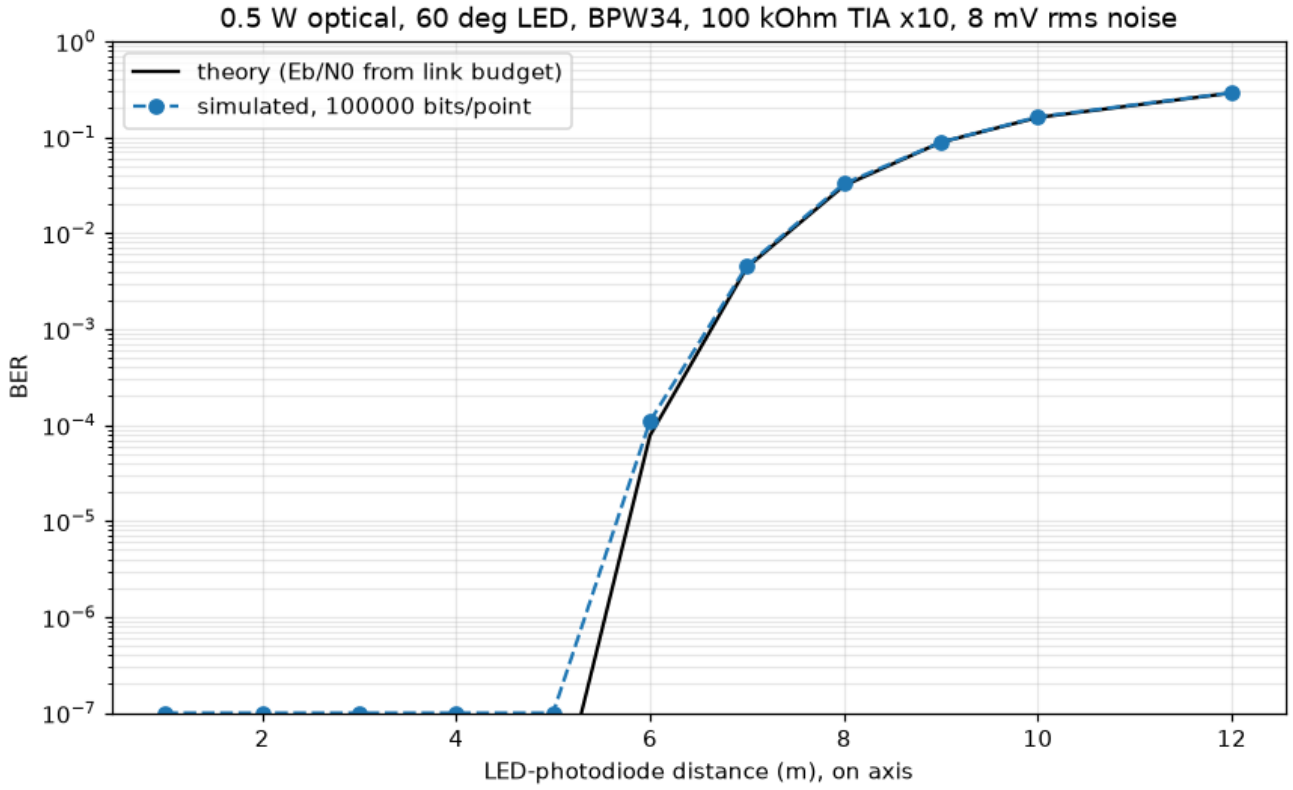


Figure 4: BER vs distance

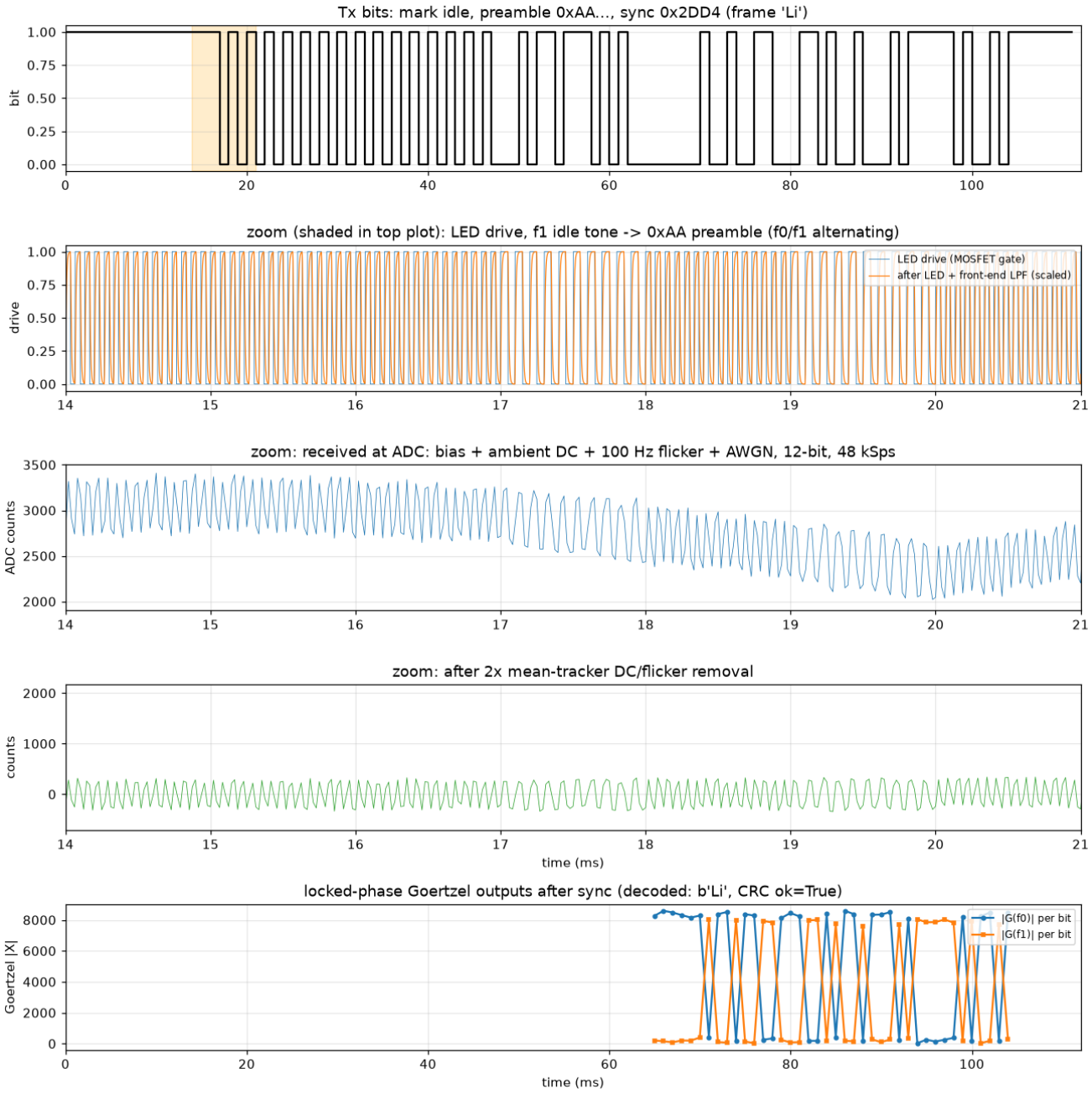


Figure 5: Waveforms